Unit 4:

Spherical Coordinate System

What We'll Cover

1. **Get to Know Spherical Coordinates:**

   Explore what the spherical coordinate system is, why it's important, and when it's useful. Understand how it provides a natural way to describe problems involving spheres, such as radiation patterns, gravitational fields, and point charges.

2. **Working with Position Vectors:**

   Learn how to express the position of a point using (r, θ, ϕ), where r is the radial distance, θ is the polar angle, and ϕ is the azimuthal angle. Relate these coordinates to rectangular coordinates (x, y, z).

What We'll Cover (continued)

3. **Coordinate Surfaces and Unit Vectors:**

   Understand surfaces of constant r, θ, and ϕ. Explore how the unit vectors $\hat{\boldsymbol{a}}_r$, $\hat{\boldsymbol{a}}_\theta$, and $\hat{\boldsymbol{a}}_{\boldsymbol{\phi}}$ vary from point to point and how they relate to the Cartesian unit vectors.

4. **Applications in Engineering and Physics:**

   See how spherical coordinates simplify problems with spherical symmetry, such as electric and magnetic fields due to point sources, sound radiation from spherical bodies, and heat flow in spherical shells.

What You'll Be Able to Do by the End of This Presentation

1. **Describe** what the spherical coordinate system is and recognize when it is the most suitable choice for describing a problem.

2. **Locate** points in space using spherical coordinates (r, θ, ϕ).

3. **Convert** between rectangular (x, y, z) and spherical (r, θ, ϕ) coordinate representations.

What You'll Be Able to Do by the End of This Presentation (continued)

4. **Visualize and interpret** the coordinate surfaces associated with constant r, θ, and ϕ.

5. **Apply** spherical coordinates to simplify and solve practical problems in physics and engineering that exhibit spherical symmetry—such as fields from point charges and radiation from spherical sources.

Introduction: The Need for Spherical Coordinates

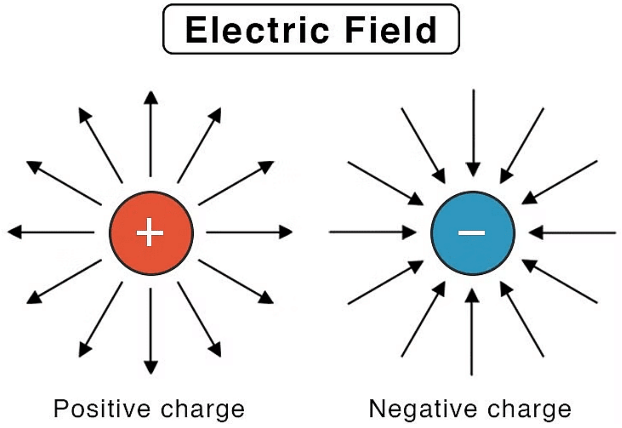
Many physical systems are not best described by straight lines or flat planes, but by **spherical symmetry**, where quantities depend only on distance from a central point and angles around it.

Examples:

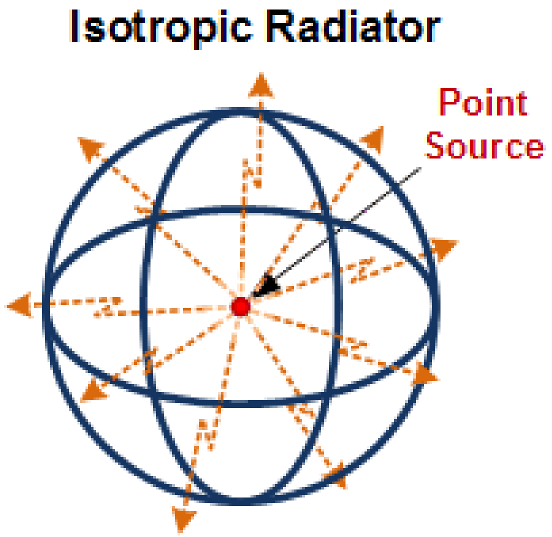
- Electric and gravitational fields around a point charge or mass.
- Radiation from a spherical antenna or isotropic source.
- Heat distribution inside a spherical shell or ball.

(See Figure 1).

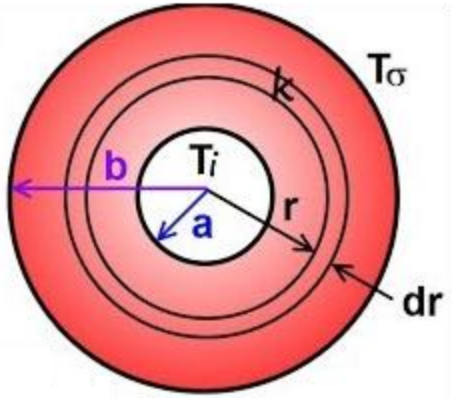
Spherical coordinates provide a **natural framework** to model and analyze these problems efficiently.



(a) Electric fields around a positive and a negative point charge.



(b) Equal power radiated in all directions.



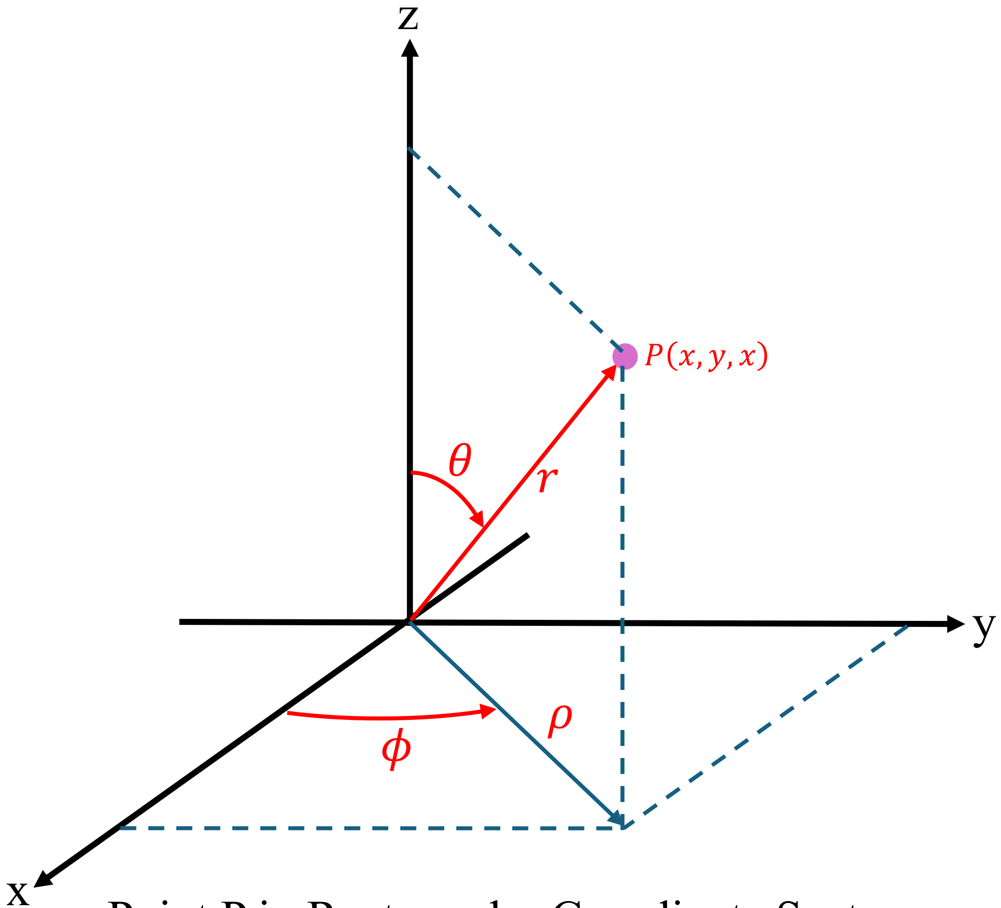
(c) Heat flow through a spherical shell.

Figure 1

The Spherical Coordinate System (Physics Convention)

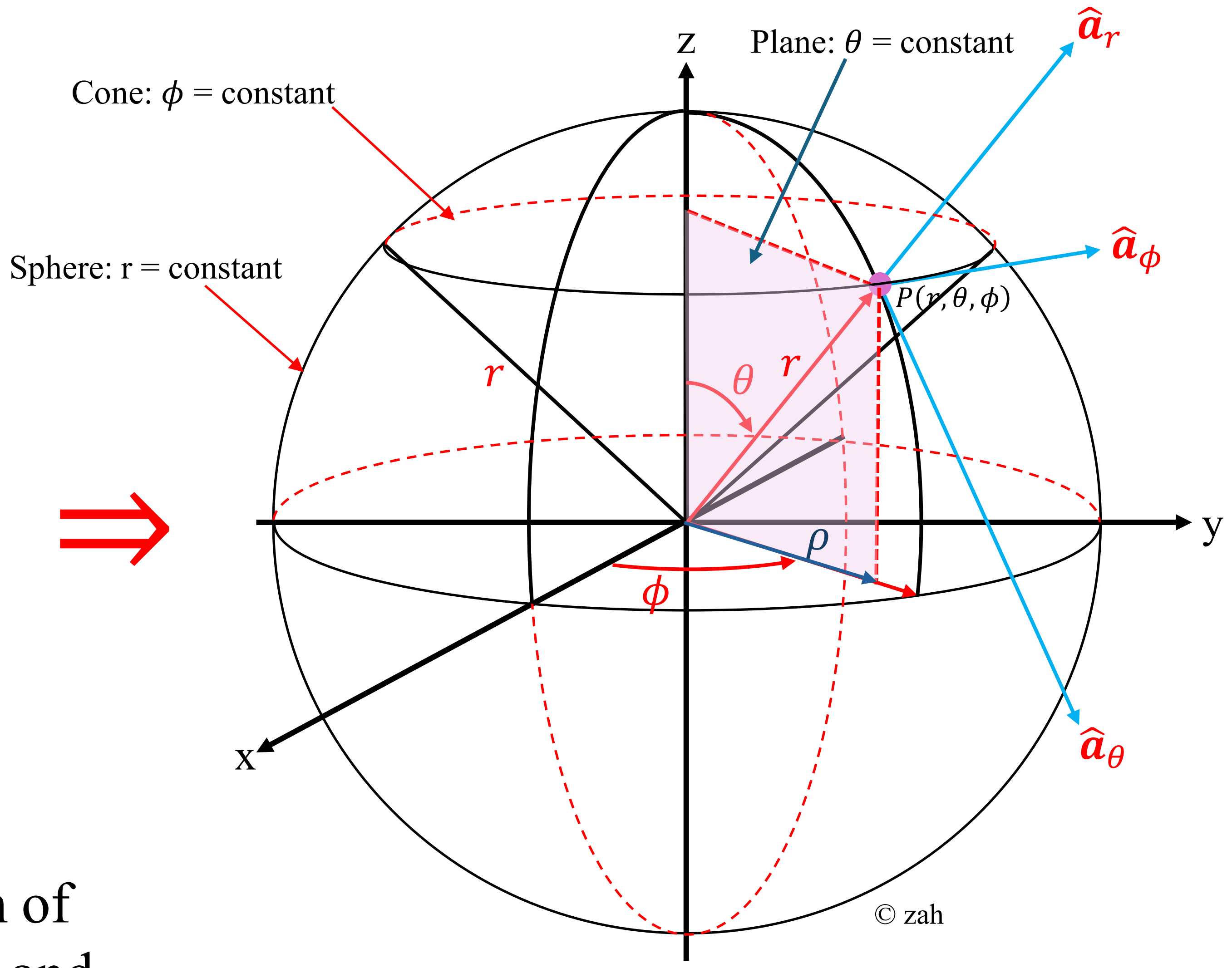
In the **spherical coordinate system**, the position of any point **P** in space — following the **physics convention** — is described by three quantities that correspond to the intersection of three mutually perpendicular coordinate surfaces (see Figure 2):

- r**:** the radial distance from the origin to the point **P**.
- ϕ **(phi):** the azimuthal angle, measured in the xy-plane from the positive x-axis to the projection of the point.
- θ **(theta):** the polar (elevation) angle, measured from the positive z-axis to the line joining the origin to the point.



Point P in Rectangular Coordinate System

$\Rightarrow$



Point P in Spherical Coordinate System

Figure 2: Geometric representation of spherical coordinate variables r, θ, and ϕ in relation to the Cartesian axes.

Hence, the coordinates of the point **P** are represented as

$$P(r, \theta, \phi)$$

where $r \geq 0$, $0 \leq \theta \leq \pi$, and $0 \leq \phi < 2\pi$.

Usage:

This convention is widely adopted in:

- **Electrical and Electronic Engineering** (e.g., antenna theory, electromagnetics)
- **Mechanical Engineering** (e.g., vibrations, dynamics)
- **Physics** (especially in electromagnetism, wave propagation, and optics)

Some Comments on Spherical Coordinate Notation

There is no single universal engineering standard for spherical coordinates, as different disciplines adopt slightly different conventions—particularly in the order and definition of the angular variables. Understanding these variations is important to avoid confusion when consulting various references. In addition to the physics convention, several other commonly used conventions are summarized below.

1. **ISO 80000-2:2019 convention**: This standard uses the following convention:

- $\boldsymbol{\rho}$ (rho): radial distance from the origin.
- $\boldsymbol{\theta}$ $(theta)$: Polar angle, measured from the positive z-axis.
- $\boldsymbol{\phi}$ (phi): azimuthal angle, measured in the xy-plane from the positive x-axis.

(see Figure 3(iii)

The coordinates are generally expressed as (ρ, θ, ϕ) with the following ranges:

$$0 \leq \rho \leq \infty; \quad 0 \leq \theta \leq \pi; \quad 0 \leq \phi \leq 2\pi$$

2. **Mathematics convention**: This standard uses the following convention:

- $\boldsymbol{r}$ **:** radial distance from the origin.
- $\boldsymbol{\theta}$ $(theta)$: Polar angle, measured from the positive z-axis.
- $\boldsymbol{\phi}$ (phi): azimuthal angle, measured in the xy-plane from the positive x-axis.

(see Figure 3(i))

The coordinates are generally expressed as (r, ϕ, θ) with the following ranges:

$$0 \leq r \leq \infty$$

$$0 \leq \phi \leq 2\pi$$

$$0 \leq \theta \leq \pi$$

3. **Wolfram Convention**: This standard defined in the Wolfram Language, uses the convention (r, θ, ϕ), where:

- $\boldsymbol{r}$ **:** The radial distance from the origin.
- $\boldsymbol{\theta}$ $(theta)$: The azimuthal angle, measured in the xy-plane from the positive x-axis.
- $\boldsymbol{\phi}$ (phi): The polar (or zenith) angle, measured from the positive z-axis.

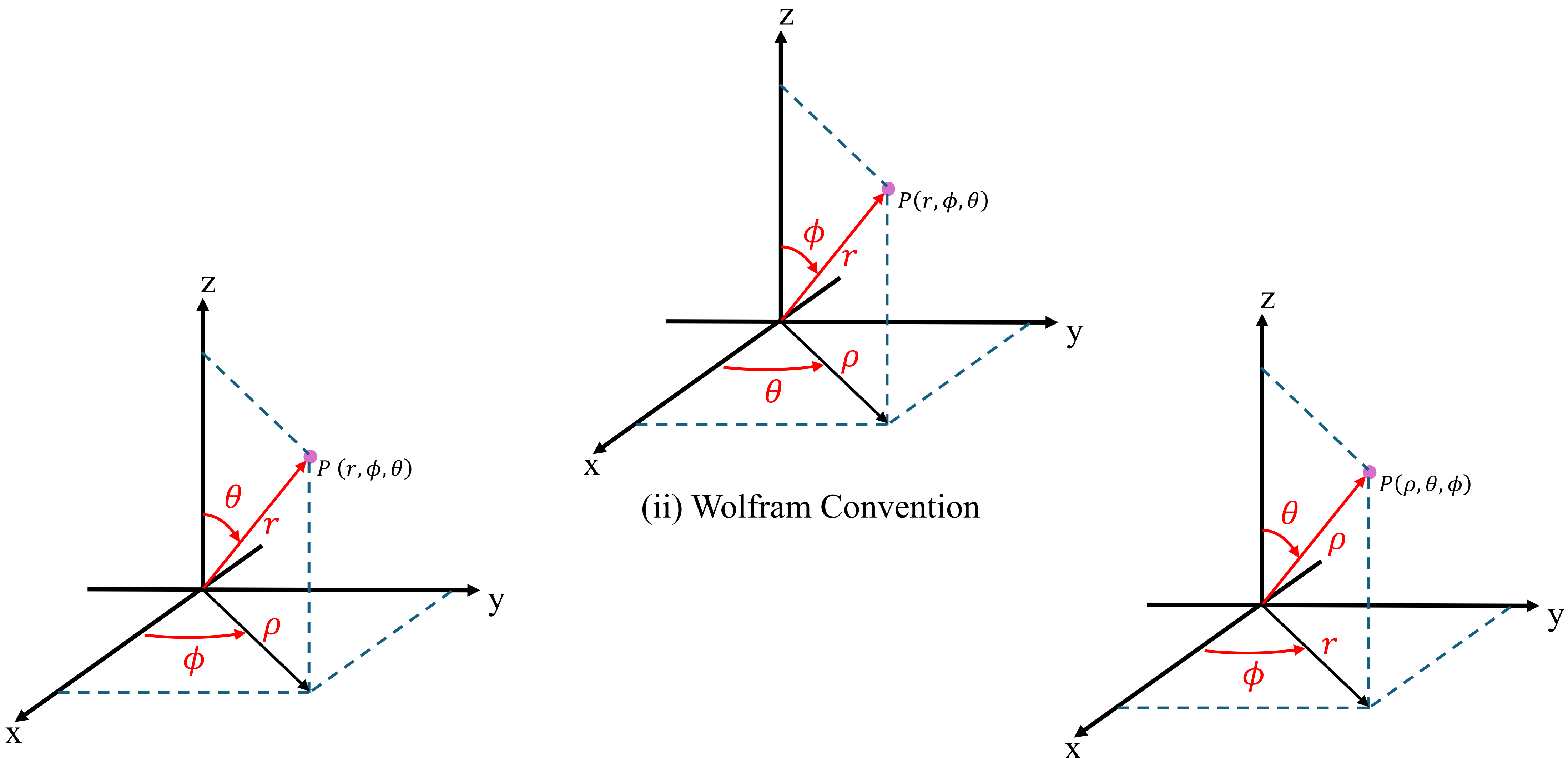
(see Figure 3(ii))

The coordinates are generally expressed as (r, ϕ, θ) and have the following ranges:

$$0 \leq r \leq \infty$$

$$0 \leq \theta \leq 2\pi$$

$$0 \leq \phi \leq \pi$$



(i) Mathematics Convention

(ii) Wolfram Convention

(iii) ISO 80000-2:2019 Convention

Figure 3: Spherical Coordinate System Conventions

Because of this difference in notation, it is important to read carefully and identify which convention is being used in any given text or reference.

Notation Used in This Course

In this course, we will use the **physics convention** for spherical coordinates to maintain clarity and consistency with standard engineering practice. The coordinates are written as:

$$(r, \theta, \phi)$$

where r is the radial distance from the origin, is the θ polar angle measured from the positive *z*-axis, and ϕ is the azimuthal angle measured from the positive *x*-axis in the *xy*-plane.

1. From Cartesian to Spherical Coordinates

A point $P(x, y, z)$ in Cartesian coordinates (see Figure 4) can be expressed in spherical coordinates (r, θ, ϕ) using the following relationships:

$$\rho = \sqrt{x^2 + y^2};\ \ r = \sqrt{x^2 + y^2 + z^2}$$

$$\phi = tan^{-1}\left(\frac{y}{x}\right)$$

$$\theta = cos^{-1}\left(\frac{z}{r}\right) = tan^{-1}\left(\frac{\sqrt{x^2+y^2}}{z}\right)$$

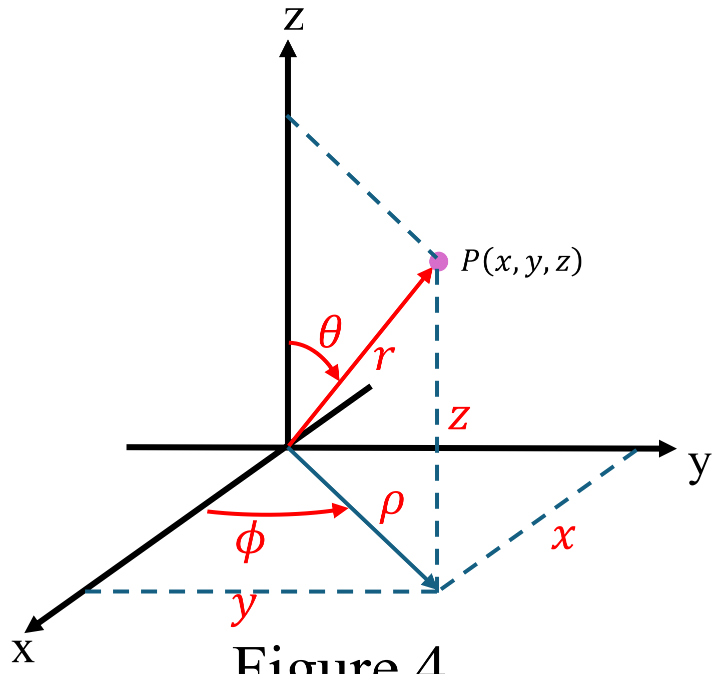


Figure 4

where:

- r = radial distance from the origin to the point,
- ϕ = azimuthal angle measured in the xy-plane from the positive x-axis,
- θ = polar (elevation) angle measured from the positive z-axis downward.

2. From Spherical to Cartesian Coordinates

Conversely, a point $P(r, \theta, \phi)$ in spherical coordinates can be expressed in Cartesian form as:

$$x = r \sin\theta \cos\phi$$

$$y = r \sin\theta \sin\phi$$

$$z = r \cos\theta$$

These equations define how the spherical coordinate surfaces map onto the three mutually perpendicular Cartesian planes (see Figure 5).

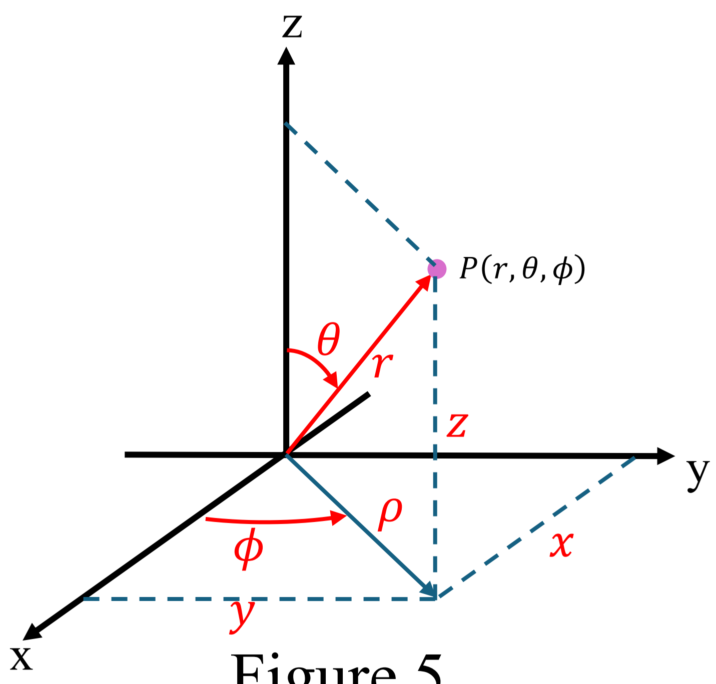


Figure 5

Worked Example 1: Cartesian → Spherical Conversion

Problem:

Convert the point

$$P(x, y, z) = (2, 4, 4)$$

from Cartesian coordinates to spherical coordinates (r, θ, ϕ), using the physics convention (see Figure 6).

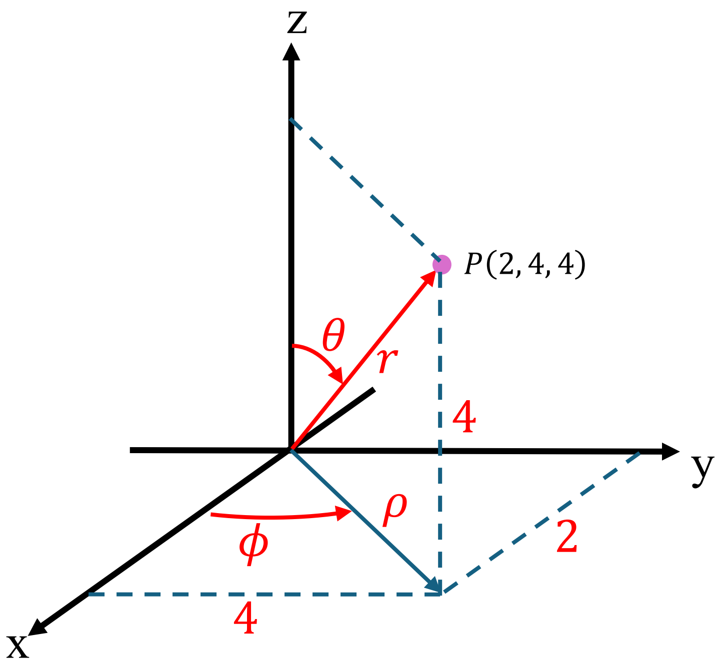


Figure 6

Step 1: Recall the Transformation Equations

From Cartesian to spherical (see Figure 7):

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\phi = tan^{-1}\left(\frac{y}{x}\right)$$

$$\theta = cos^{-1}\left(\frac{z}{r}\right) = tan^{-1}\left(\frac{\sqrt{x^2+y^2}}{z}\right)$$

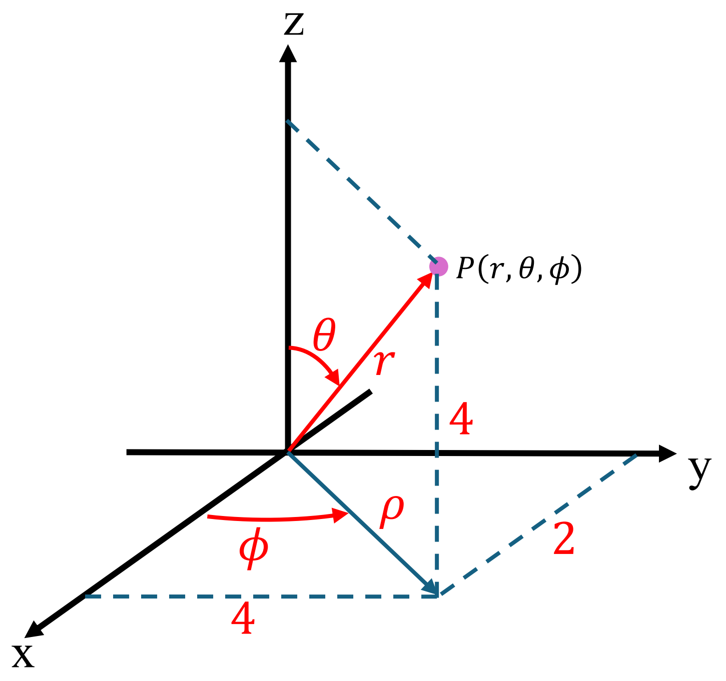


Figure 7

Step 2: Substitute the Given Values

$$r = \sqrt{2^2 + 4^2 + 4^2} = \sqrt{36} = 6.0$$

$$\phi = tan^{-1}\left(\frac{4}{2}\right) = 63.435°$$

$$\theta = cos^{-1}\left(\frac{4}{6}\right) = 33.55°$$

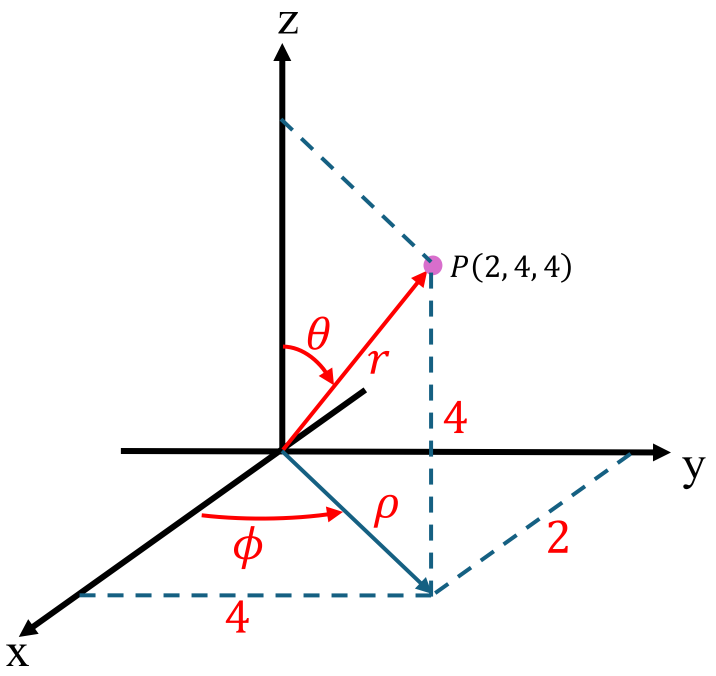


Figure 6

Step 3: Express the Final Answer

$$P(r, \theta, \phi) = (6.0, 63.435°, 33.55°)$$

Thus, the point P can be visualized as lying in the first octant, somewhere between the z-axis and the xy-plane, as shown in Figure 8.

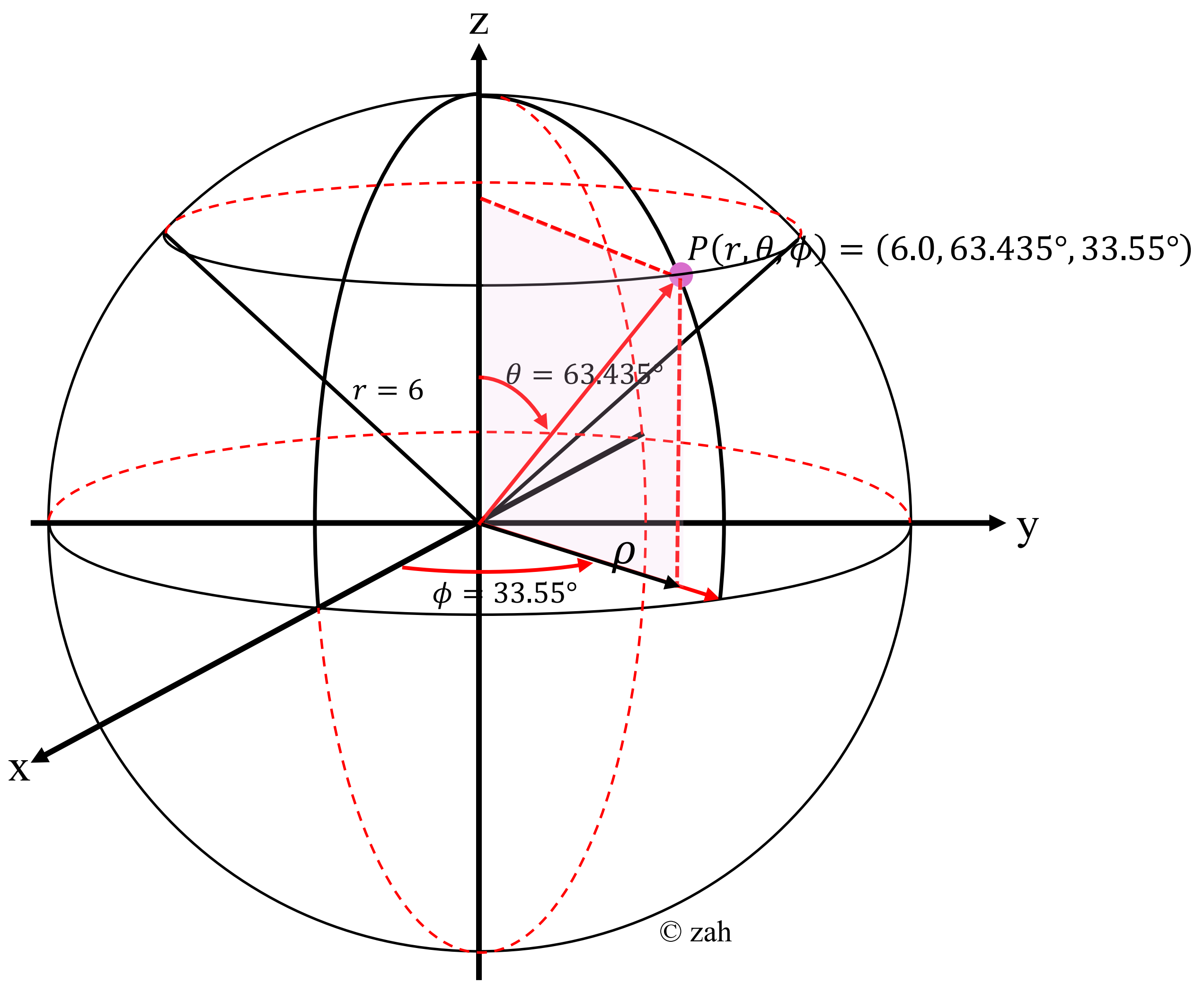


Figure 8: Location of point $P(r, \theta, \phi) = (6.0, 63.435°, 33.55°)$ in the spherical coordinate system.

Position Vector in the Spherical Coordinate System

In the spherical coordinate system (see Figure 9), the position of a point P is specified by three coordinates:

- r: the radial distance from the origin O to the point P,
- θ: the polar angle measured from the positive z-axis, and
- ϕ: the azimuthal angle measured in the xy-plane from the positive x-axis.

Thus, the coordinates of point P are written in the order (r, θ, ϕ).

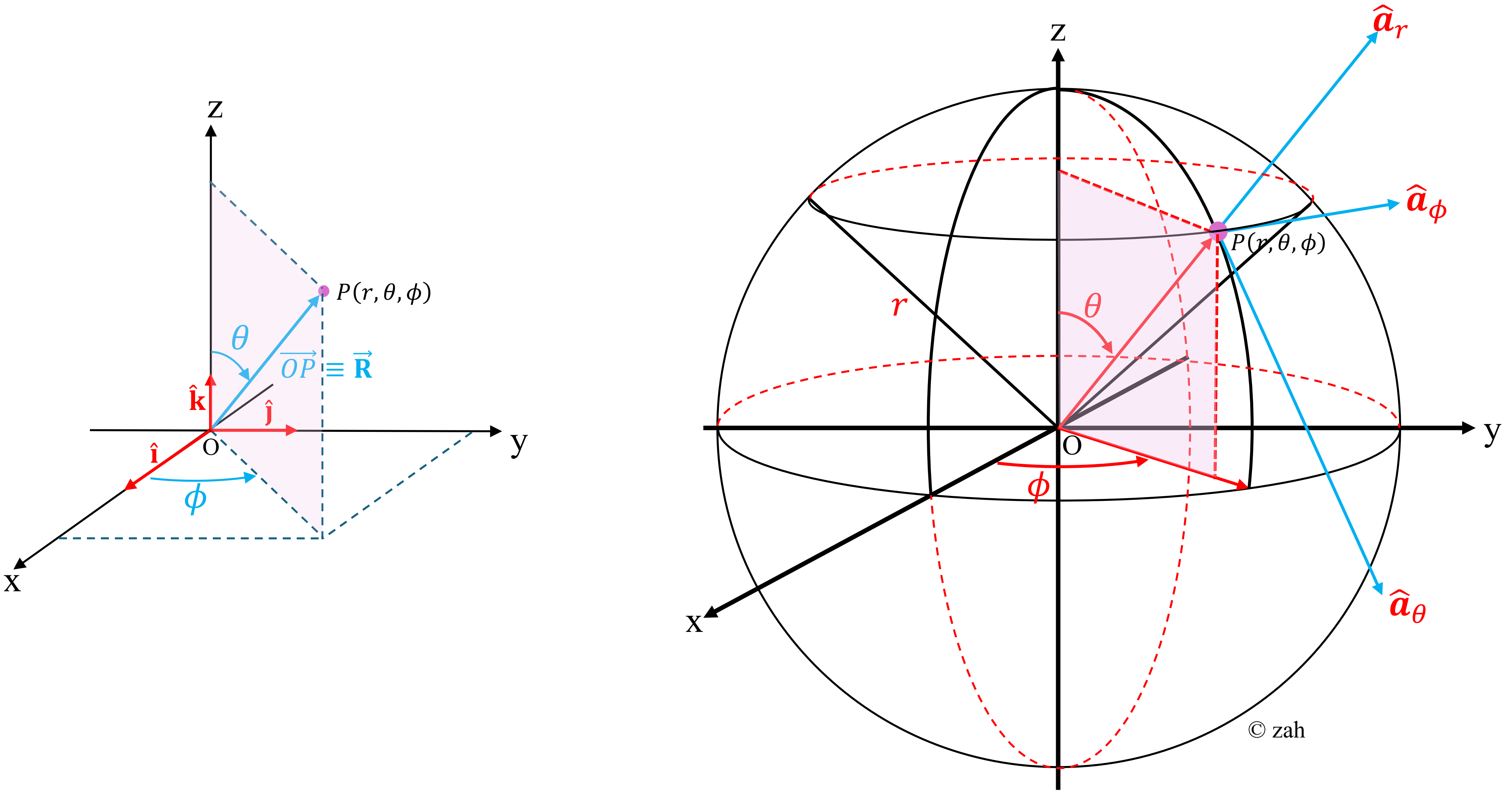


Figure 9: Position vector $\overrightarrow{OP}$ in the spherical coordinate system.

The **position vector** of point P, denoted by $\overrightarrow{OP}$ or $\vec{\mathbf{R}}(x, y, z)$, can be expressed in Cartesian unit vectors $\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}}$ as:

$$\begin{aligned}\overrightarrow{OP} \equiv \vec{\mathbf{R}}(x, y, z) &= r\sin\theta\cos\phi\,\hat{\mathbf{i}} + r\sin\theta\sin\phi\,\hat{\mathbf{j}} + r\cos\theta\,\hat{\mathbf{k}} \\ &= r\left(\sin\theta\cos\phi\,\hat{\mathbf{i}} + \sin\theta\sin\phi\,\hat{\mathbf{j}} + \cos\theta\,\hat{\mathbf{k}}\right)\end{aligned}$$

In **column vector form**, this can be written as:

$$\vec{\mathbf{R}}(x, y, z) = \begin{bmatrix} r \sin\theta \cos\phi \\ r \sin\theta \sin\phi \\ r \cos\theta \end{bmatrix} = r \begin{bmatrix} \sin\theta \cos\phi \\ \sin\theta \sin\phi \\ \cos\theta \end{bmatrix} = r\, \hat{\mathbf{r}}(\theta, \phi)$$

where the **unit vector in the radial direction** is given by:

$$\hat{\mathbf{r}}(\theta, \phi) = \begin{bmatrix} \sin\theta \cos\phi \\ \sin\theta \sin\phi \\ \cos\theta \end{bmatrix}$$

Note

In this context, the symbol $\hat{\mathbf{R}}$ is used to emphasize that it represents the **unit vector in the direction of R**. That is,

$$\hat{\mathbf{R}} = \frac{\mathbf{R}}{|\,\mathbf{R}\,|} = \frac{\mathbf{R}}{r}$$

Alternatively, some texts or authors use $\hat{\mathbf{a}}_r$ to denote a **unit vector pointing radially outward**, i.e., in the same direction as **R**. Hence, the same vector can also be written as

$$\mathbf{R} = r\,\hat{\mathbf{a}}_r$$

where:

- $r = \sqrt{x^2 + y^2 + z^2}$ is the **magnitude (distance)** from the origin,
- $\hat{\mathbf{a}}_r$ is the **radial unit vector**, pointing outward from the origin in the direction of **R**.

Worked Example 2

Given the position vector in rectangular coordinates:

$$\vec{\mathbf{R}}(x, y, z) = \langle 3, 4, 5 \rangle \text{ m}$$

(see Figure 10) express it in spherical coordinates as:

$$\vec{\mathbf{R}}(x, y, z) = r\,\hat{\mathbf{r}}(\theta, \phi)$$

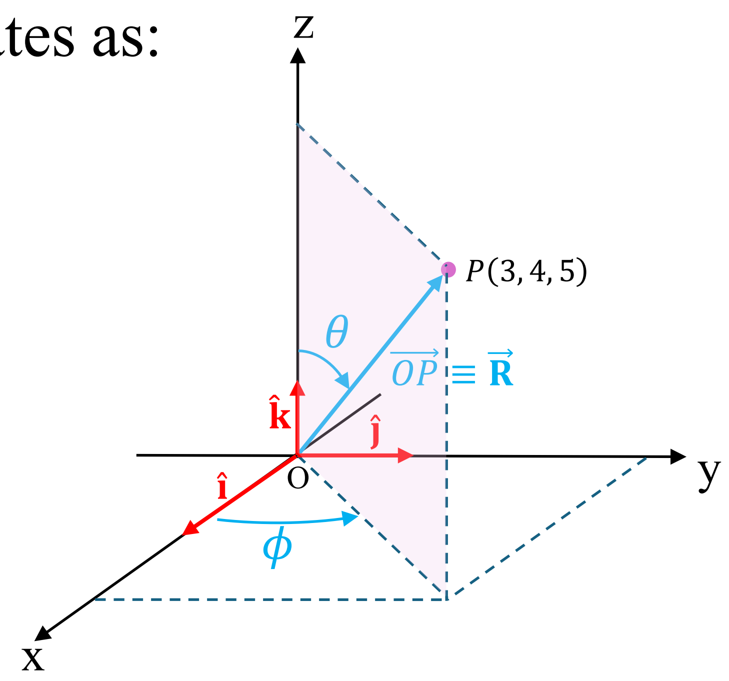


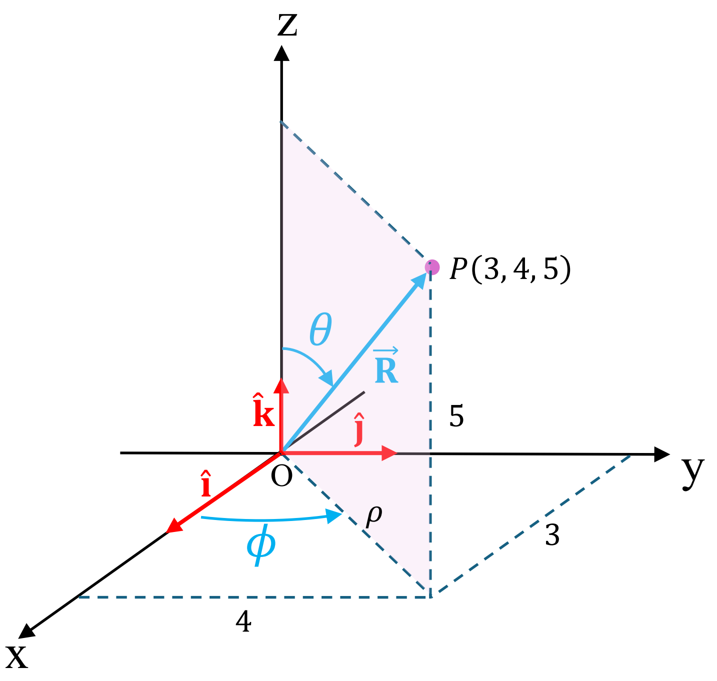
Figure 10

Solution

Step 1: Compute the radial distance r (see Figure 11)

$$\rho = \sqrt{x^2 + y^2} = \sqrt{(3)^2 + (4)^2} = \sqrt{25} = 5$$

$$r = \sqrt{\rho^2 + z^2} = \sqrt{5^2 + 5^2} = \sqrt{50} = 5\sqrt{2}\ \mathrm{m}$$



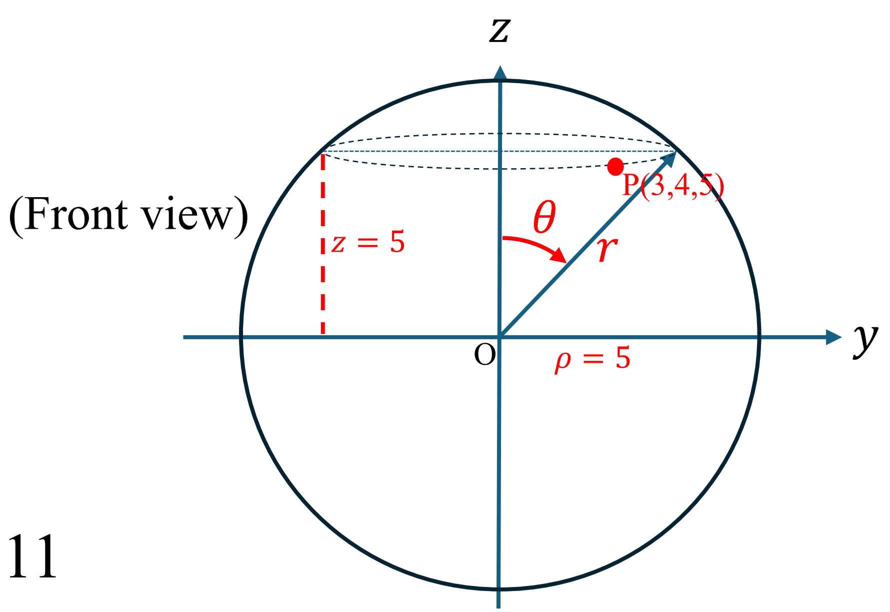


Figure 11

Step 2: Determine the polar angle θ (see Figure 12)

The polar angle θ is measured from the **positive z-axis**:

$$\theta = cos^{-1}\left(\frac{z}{r}\right)$$

$$\theta = cos^{-1}\left(\frac{5}{5\sqrt{2}}\right) = cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = 45°$$

(Front view)

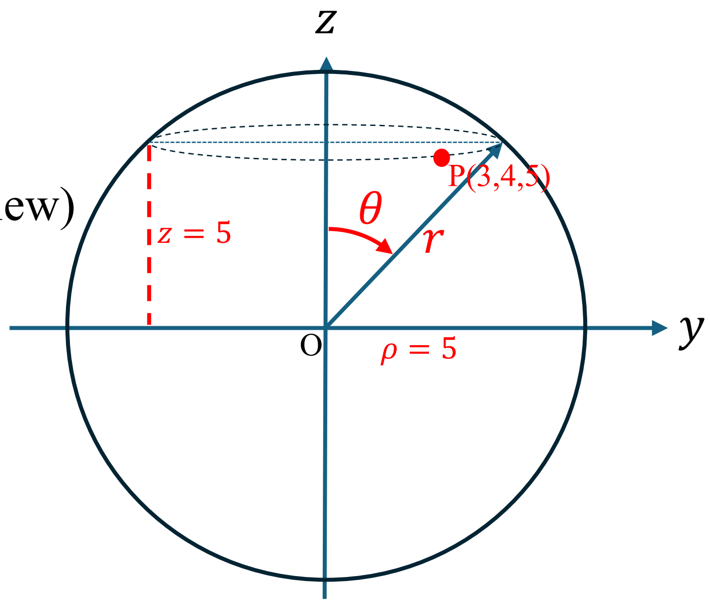


Figure 12

Step 3: Determine the azimuthal angle ϕ

The azimuthal angle ϕ is measured in the **xy-plane** <u>from the positive x-axis</u>:

$$\phi = tan^{-1}\left(\frac{y}{x}\right)$$

$$\phi = tan^{-1}\left(\frac{4}{3}\right) = 53.13°$$

(see Figure 13)

(Top view)

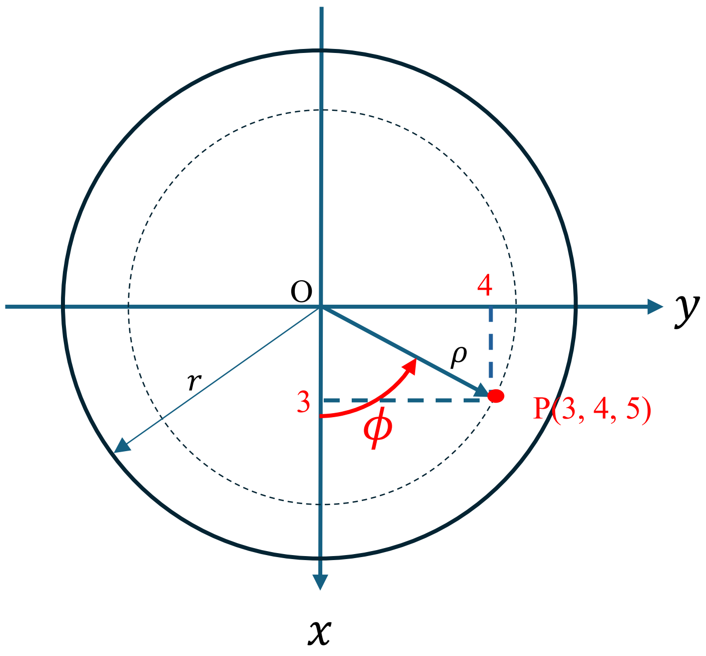


Figure 13

Final Answer

The position vector in spherical coordinates is:

$$\vec{\mathbf{R}} = 5\sqrt{2}\,\hat{\mathbf{r}}(45°, 53.13°)$$

where

$$\hat{\mathbf{r}}(45°, 53.13°) = \begin{bmatrix} \sin(45°)\cos(53.13°) \\ \sin(45°)\sin(53.13°) \\ \cos(45°) \end{bmatrix} = \begin{bmatrix} 0.424 \\ 0.566 \\ 0.707 \end{bmatrix}$$

Worked Example 3

Problem

Given the position vector in rectangular coordinates:

$$\vec{\mathbf{R}}(x, y, z) = \langle 1, -4, -5 \rangle \text{ m}$$

express the vector in **spherical coordinates** (see Figure 14).

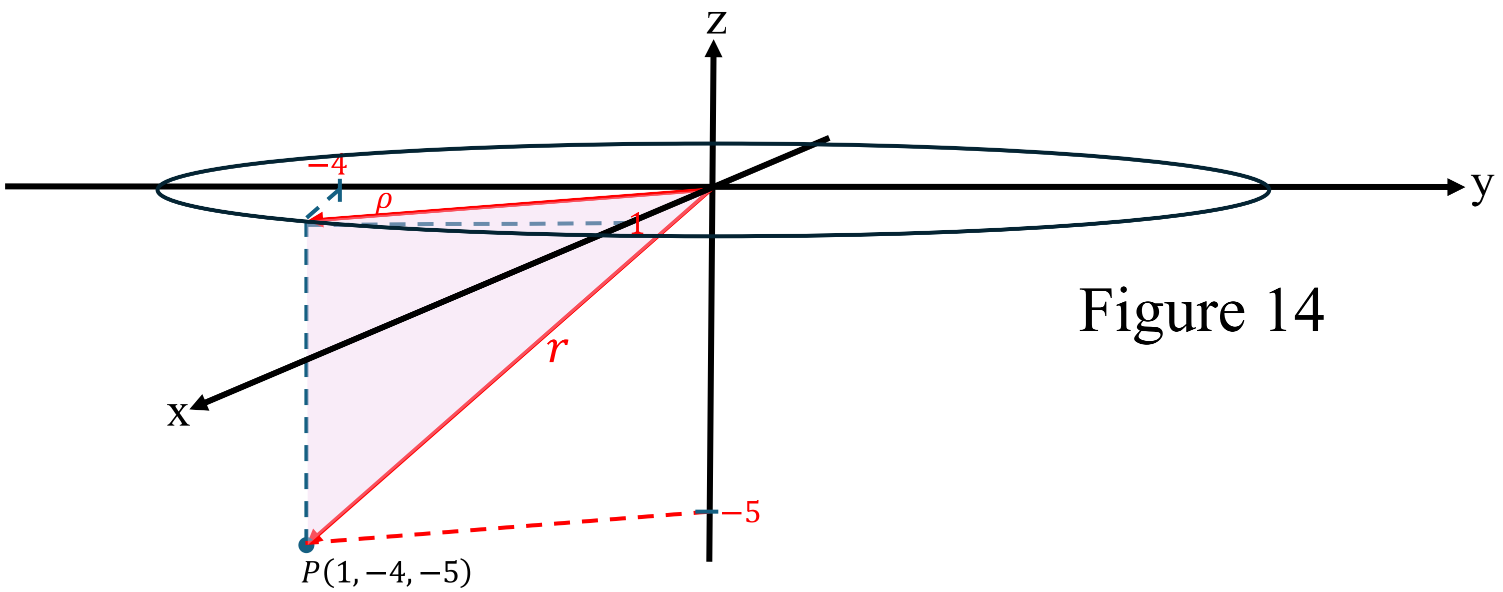


Figure 14

Solution

Step 1: Recall the conversion relations

For any point (x, y, z) in rectangular coordinates, the corresponding spherical coordinates are defined as:

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\phi = tan^{-1}\left(\frac{y}{x}\right)$$

$$\theta = cos^{-1}\left(\frac{z}{r}\right)$$

where:

- r = radial distance from the origin
- ϕ = azimuthal angle measured in the xy-plane from the +x axis
- θ = polar angle measured from the +z axis

Step 2: Substitute the given rectangular coordinates

$$x = 1, \qquad y = -4, \qquad z = -5$$

Compute each spherical coordinate.

(a) Radial distance r

$$\rho = \sqrt{x^2 + y^2} = \sqrt{(1)^2 + (-4)^2} = \sqrt{17}$$

$$r = \sqrt{\rho^2 + z^2}$$

$$r = \sqrt{17 + 5^2} = \sqrt{42}$$

$$r = \sqrt{42} = 6.48 \text{ m}$$

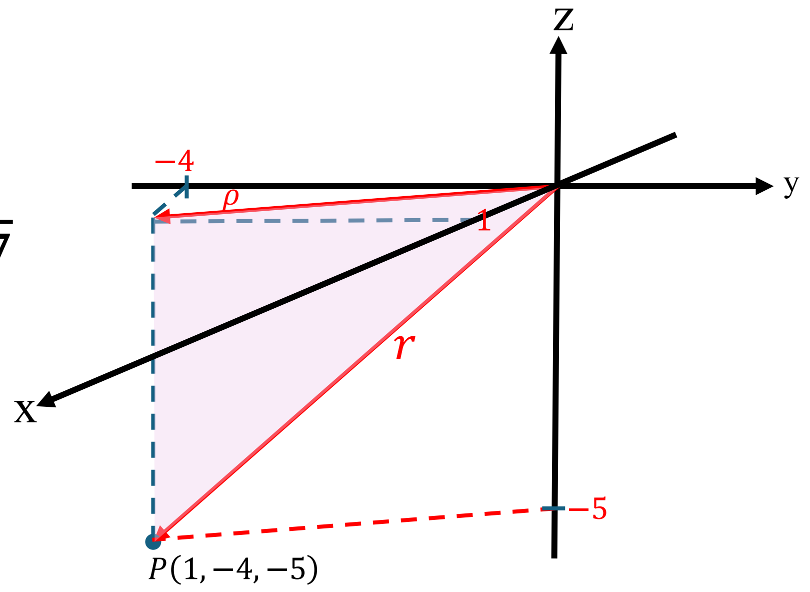


Figure 14

(b) Azimuthal angle ϕ

$$\phi = tan^{-1}\left(\frac{y}{x}\right) = tan^{-1}\left(\frac{-4}{1}\right)$$

$$\phi = tan^{-1}(-4) = -75.96°$$

Since the point lies in the **fourth quadrant** ($x > 0$, $y < 0$),

$$\phi = 360° - 75.96° = 284.04°$$

(see Figure 15)

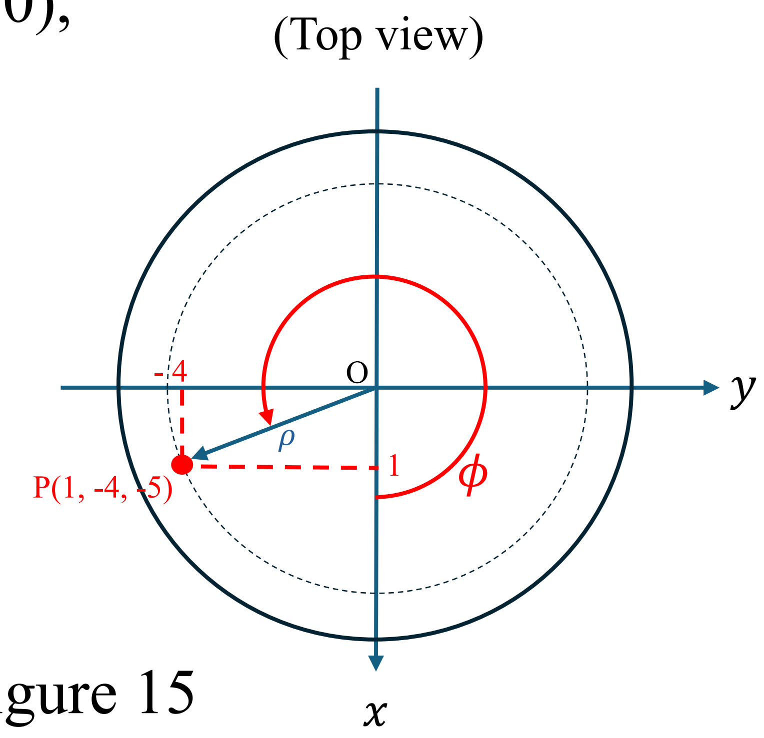


Figure 15

(c) Polar angle θ

$$\theta = cos^{-1}\left(\frac{z}{r}\right) = cos^{-1}\left(\frac{-5}{6.48}\right)$$

$$\theta = cos^{-1}(-0.7716) = -140.52°$$

(see Figure 16)

(Top view)

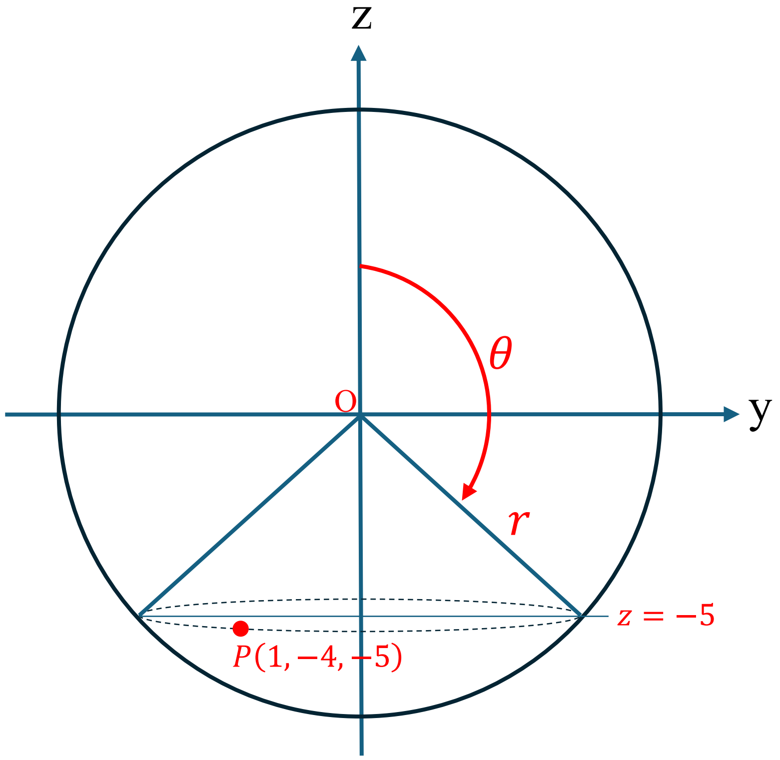


Figure 16

(c) Polar angle θ (Alternative solution steps)

$$\theta' = cos^{-1}\left(\frac{z}{r}\right) = cos^{-1}\left(\frac{5}{6.48}\right)$$

$$\theta' = cos^{-1}(0.7716) = 39.502°$$

$$\theta = 180° - \theta' = 180 - 39.502 = 140.5°$$

(see Figure 17)

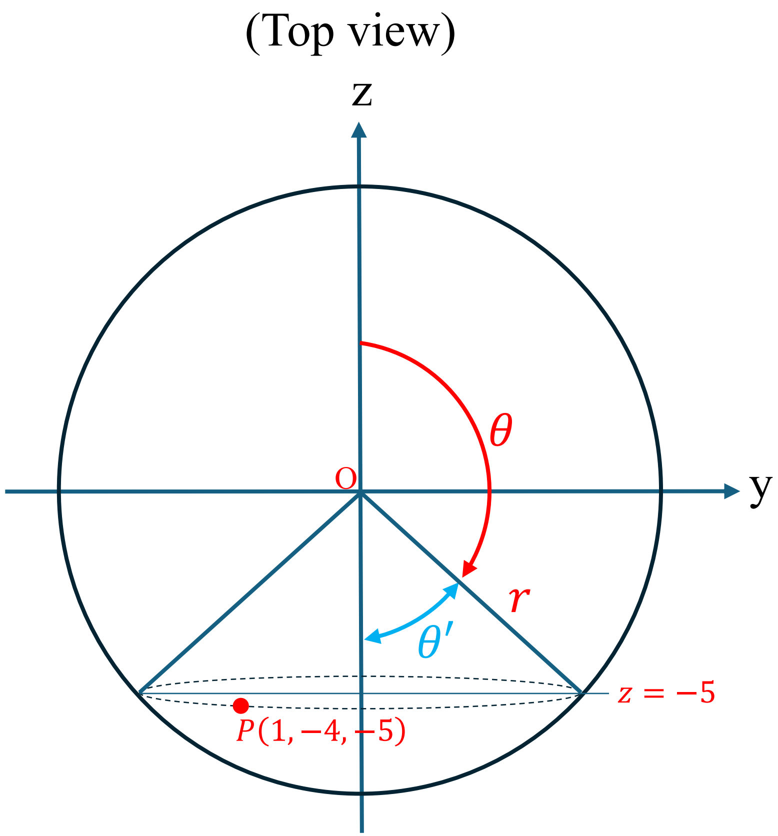


Figure 17

Step 3: Write the spherical coordinates

Hence, the point P in spherical coordinates is:

$$(r, \theta, \phi) = (6.48\ \mathrm{m}, 140.52°, 284.04°)$$

Step 4: Verification

Check by converting back to rectangular form:

$$x = r\sin\theta\cos\phi = \sqrt{42}\sin(140.52°)\cos(284.04°) \approx 1$$

$$y = r\sin\theta\sin\phi = \sqrt{42}\sin(140.52°)\sin(284.04°) \approx -4$$

$$z = r\cos\theta = \sqrt{42}\cos(140.52°) \approx -5$$

Verified.

Step 5: Express the position vector in spherical form

The position vector can now be written as:

$$\overrightarrow{\mathbf{R}} = r\,\hat{\mathbf{r}} = 6.48\,\hat{\mathbf{r}}$$

and the corresponding **spherical coordinates** of the point are:

$$(r, \theta, \phi) = (6.48\ \mathrm{m}, 140.52°, 284.04°\ )$$

Final Answer

$$\overrightarrow{\mathbf{R}} = 6.48\ \hat{\mathbf{r}}$$

$$(r, \theta, \phi) = (6.48\ \mathrm{m}, 140.52°, 284.04°\ )$$

Converting Physical Vectors Between Rectangular and Spherical Coordinates

In engineering analysis, it is often necessary to express a physical vector—such as an electric field, or a magnetic field—in a coordinate system that best matches the geometry of the problem. Many electromagnetic and field-related problems exhibit **radial symmetry**, where the **spherical coordinate system** provides a more natural and convenient representation than the rectangular (Cartesian) system. By transforming vectors between these coordinate systems, engineers can simplify mathematical expressions and better interpret the underlying physical behavior.

This section explains how to convert a physical vector between rectangular and spherical coordinates using **matrix transformation relations**, followed by worked numerical examples.

A physical vector **A** in the spherical coordinate system may be written as

$$\mathbf{A} = \langle A_r, A_\theta, A_\phi \rangle$$

or

$$\mathbf{A} = A_r \boldsymbol{a}_r + A_\theta \boldsymbol{a}_\theta + A_\phi \boldsymbol{a}_\phi$$

where $\boldsymbol{a}_r$, $\boldsymbol{a}_\theta$, and $\boldsymbol{a}_\phi$ are unit vectors along the r-, θ-, and ϕ-directions, respectively (see Figure 18).

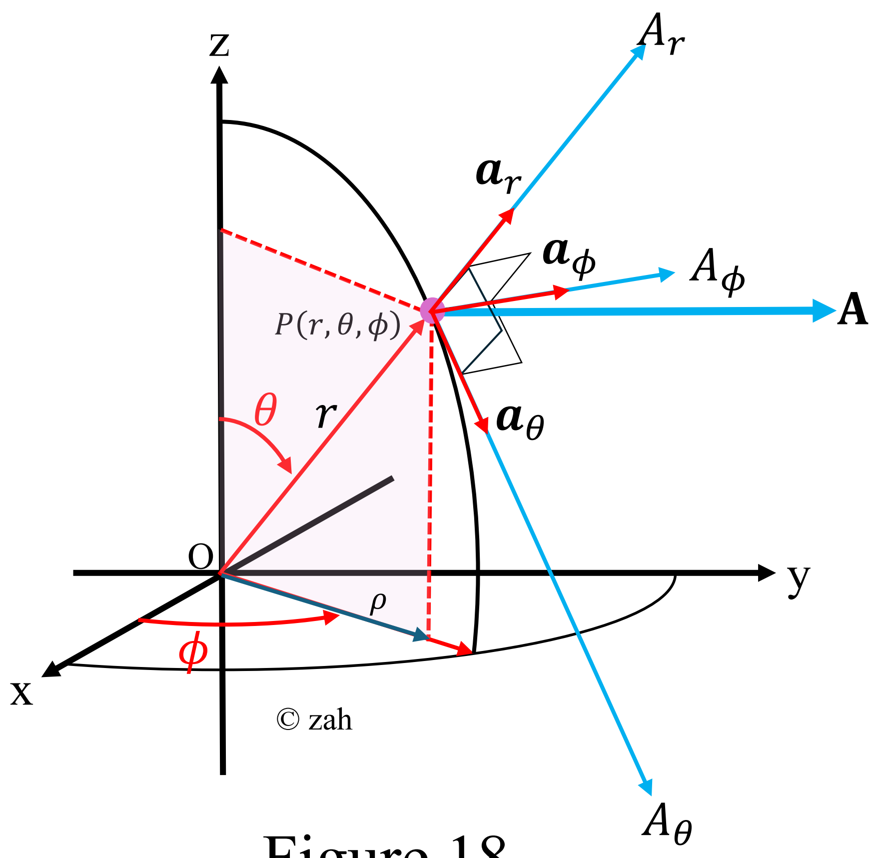


Figure 18

The units $\boldsymbol{a}_r$, $\boldsymbol{a}_\theta$, and $\boldsymbol{a}_\phi$ are mutually orthogonal; $\boldsymbol{a}_r$ being directed along the radius or in the direction of increasing r, $\boldsymbol{a}_\theta$ in the direction of increasing θ, and $\boldsymbol{a}_\theta$ in the direction of increasing ϕ (see Figure 18). Thus,

$$\boldsymbol{a}_r \cdot \boldsymbol{a}_r = \boldsymbol{a}_\theta \cdot \boldsymbol{a}_\theta = \boldsymbol{a}_\phi \cdot \boldsymbol{a}_\phi = 1$$

$$\boldsymbol{a}_r \cdot \boldsymbol{a}_\theta = \boldsymbol{a}_\theta \cdot \boldsymbol{a}_\phi = \boldsymbol{a}_\phi \cdot \boldsymbol{a}_r = 0$$

$$\boldsymbol{a}_r \times \boldsymbol{a}_\theta = \boldsymbol{a}_\phi$$

$$\boldsymbol{a}_\theta \times \boldsymbol{a}_\phi = \boldsymbol{a}_r$$

$$\boldsymbol{a}_\phi \times \boldsymbol{a}_r = \boldsymbol{a}_\theta$$

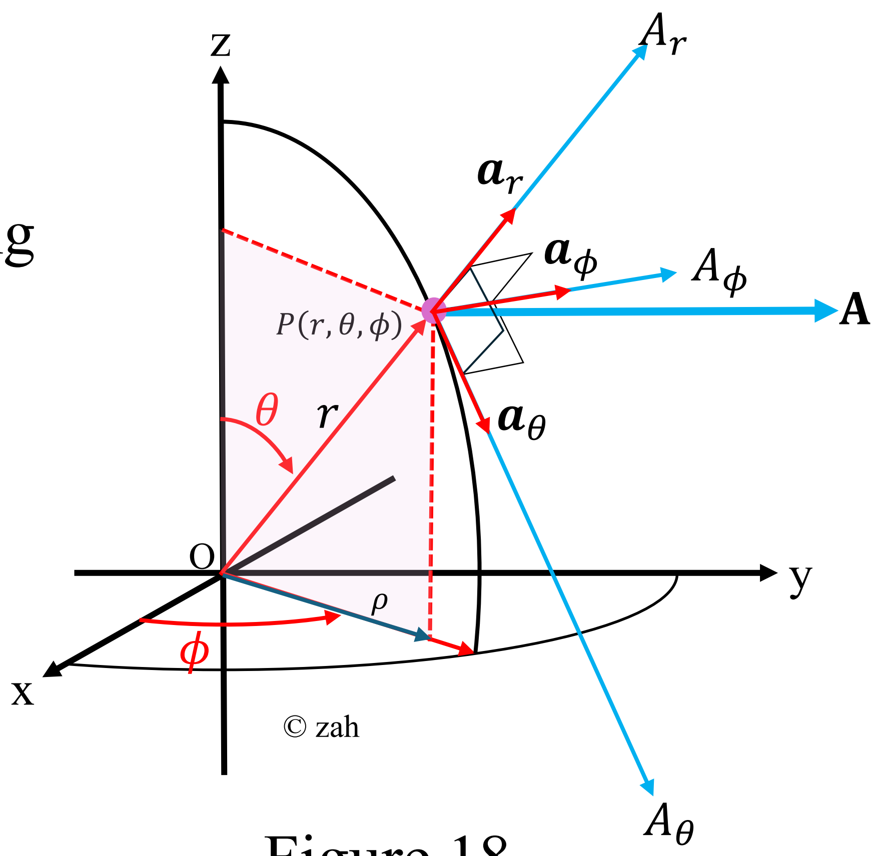


Figure 18

The magnitude of **A** is

$$|\mathbf{A}| = \left(A_r^2 + A_\theta^2 + A_\phi^2\right)^{1/2}$$

(see Figure 19)

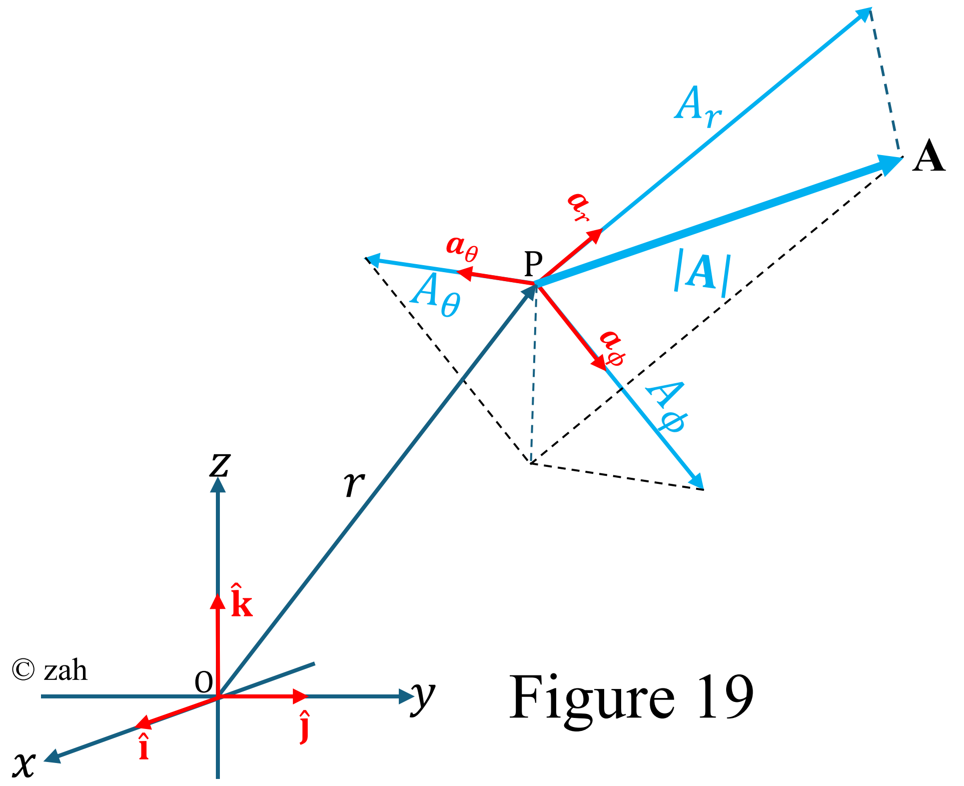


Figure 19

Note

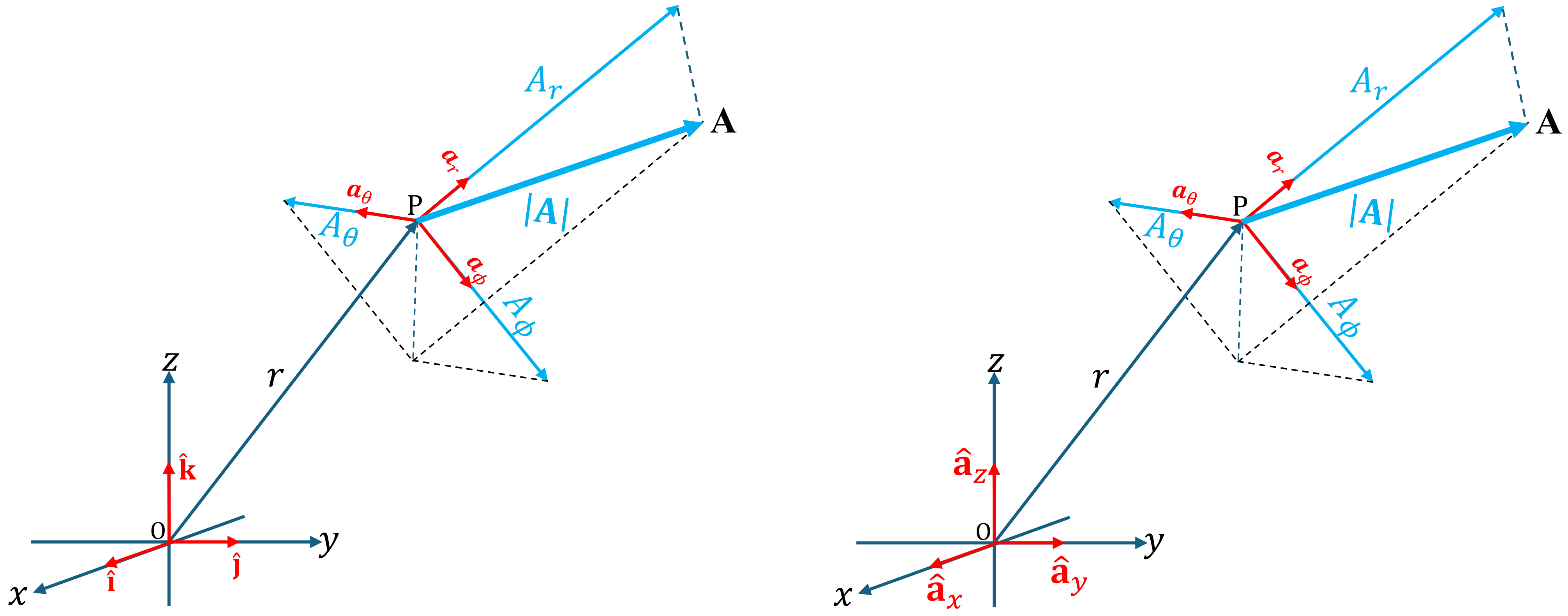
Some literature adopts an alternative notation for the rectangular unit vectors, using $\hat{\mathbf{a}}_x$, $\hat{\mathbf{a}}_y$, $\hat{\mathbf{a}}_z$ in place of the more common **î, ĵ, k̂** symbols. Thus, instead of writing a physical vector **A** as

$$\mathbf{A} = A_x\,\hat{\mathbf{i}} + A_y\,\hat{\mathbf{j}} + A_z\,\hat{\mathbf{k}},$$

it may alternatively be expressed as

$$\mathbf{A} = A_x\,\mathbf{a}_x + A_y\,\mathbf{a}_y + A_z\,\mathbf{a}_z.$$

Both notations are equivalent and represent the same vector components along the **x**, **y**, and **z** axes, respectively (see Figure 20). The choice of notation typically depends on the convention used in a particular textbook or discipline.



(i) $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, $\hat{\mathbf{k}}$ notation

(ii) $\hat{\mathbf{a}}_x$, $\hat{\mathbf{a}}_y$, $\hat{\mathbf{a}}_z$ notation

Figure 20

Vector Component Transformation

Let a vector **A** be expressed in rectangular form:

$$\mathbf{A}(x, y, z) = \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

and in spherical form:

$$\mathbf{A}(r, \theta, \phi) = \begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix}.$$

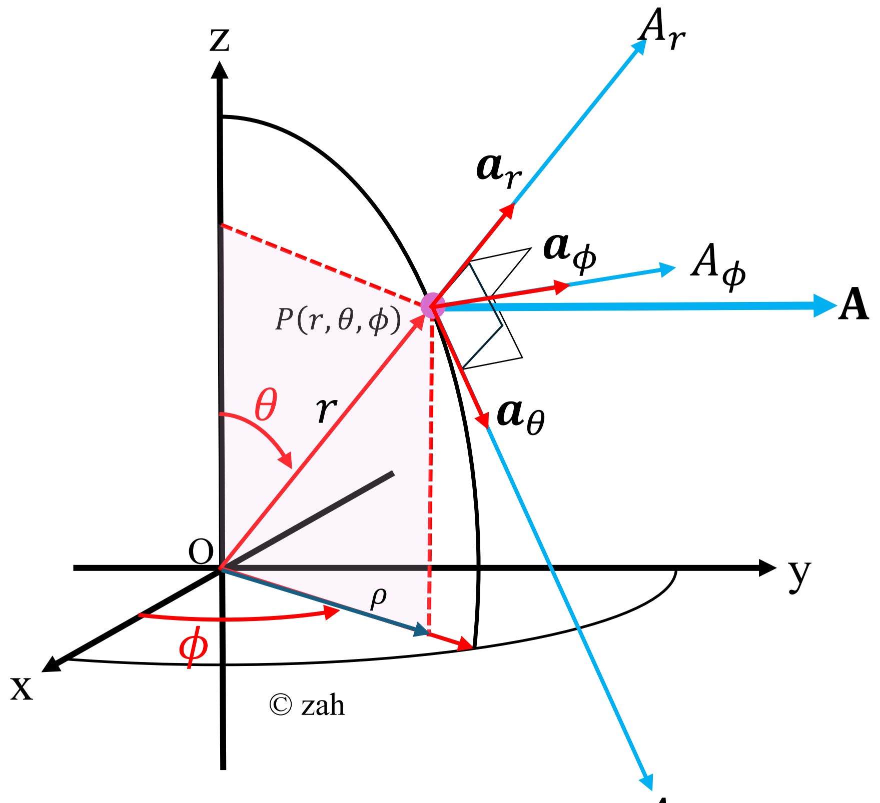


Figure 18

The transformation from rectangular to spherical coordinates is computed using the relation

$$\mathbf{A}(r,\theta,\phi) = [T]\mathbf{A}(x,y,z)$$

or

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = [T]\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

where

$$[T] = \begin{bmatrix} \sin\theta\cos\phi & \sin\theta\sin\phi & \cos\theta \\ \cos\theta\cos\phi & \cos\theta\sin\phi & -\sin\theta \\ -\sin\phi & \cos\phi & 0 \end{bmatrix}$$

is the transformation matrix.

Conversely, the transformation from spherical to rectangular coordinates is computed using the relation

$$\mathbf{A}(x, y, z) = [T]^{\mathrm{T}}\mathbf{A}(r, \theta, \phi)$$

or

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = [T]^{\mathrm{T}} \begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix}$$

where is $[T]^{\mathrm{T}}$ is the transpose of matrix T and

$$[T]^{\mathrm{T}} = \begin{bmatrix} \sin\theta\cos\phi & \cos\theta\cos\phi & -\sin\phi \\ \sin\theta\sin\phi & \cos\theta\sin\phi & \cos\phi \\ cos\,\theta & -\sin\theta & 0 \end{bmatrix}$$

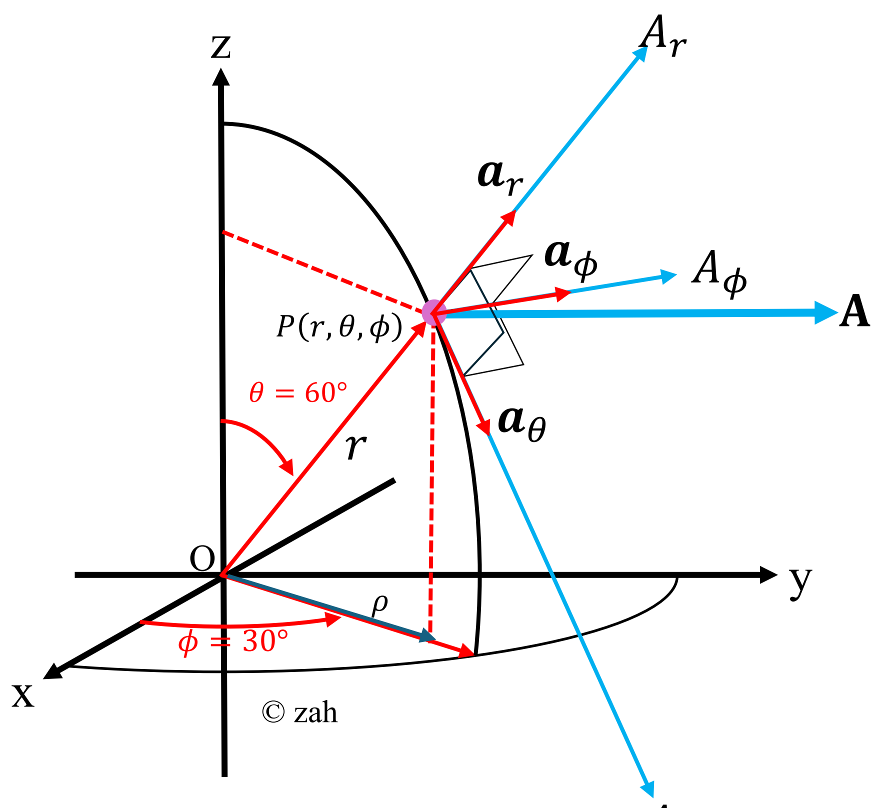
Worked Example 4

Given a physical vector $\mathbf{A} = 3\hat{\boldsymbol{a}}_x + 4\hat{\boldsymbol{a}}_y + 2\hat{\boldsymbol{a}}_z$.

Evaluate the spherical components A_r, A_θ, A_ϕ of $\mathbf{A}$ at the point with spherical angles

$$\theta = 60°, \phi = 30°$$

(see Figure 21)



z
A_r
$\boldsymbol{a}_r$
$\boldsymbol{a}_\phi$
A_ϕ
$\mathbf{A}$
$P(r, \theta, \phi)$
$\theta = 60°$
r
$\boldsymbol{a}_\theta$
O
y
ρ
$\phi = 30°$
x
© zah
A_θ

Figure 21

Solution

1. Compute Transformation Matrix [T] Use: $\theta = 60°, \quad \phi = 30°$

$$[T] = \begin{bmatrix} \sin\theta\cos\phi & \sin\theta\sin\phi & \cos\theta \\ \cos\theta\cos\phi & \cos\theta\sin\phi & -\sin\theta \\ -\sin\phi & \cos\phi & 0 \end{bmatrix}$$

$$[T] = \begin{bmatrix} \sin(60°)\cos(53.13°) & \sin(60°)\sin(53.13°) & \cos(60°) \\ \cos(60°)\cos(53.13°) & \cos(60°)\sin(53.13°) & -\sin(60°) \\ -\sin(30°) & \cos(30°) & 0 \end{bmatrix}$$

$$[T] = \begin{bmatrix} 0.75 & 0.433 & 0.5 \\ 0.433 & 0.25 & -0.866 \\ -0.5 & 0.866 & 0 \end{bmatrix}$$

2. Compute spherical components by dot products

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = [T] \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} 0.75 & 0.433 & 0.5 \\ 0.433 & 0.25 & -0.866 \\ -0.5 & 0.866 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 4.982 \\ 0.567 \\ 1.964 \end{bmatrix}$$

Hence, the spherical components of the vector **A** are:

$$A_r = 4.982,\ A_\theta = 0.567,\ A_\phi = 1.964$$

3. Express the vector in the spherical basis

Therefore at the chosen point,

$$\mathbf{A}(r,\theta,\phi) = A_r\ \hat{\boldsymbol{a}}_{\boldsymbol{r}} + A_\theta\ \hat{\boldsymbol{a}}_{\boldsymbol{\theta}} + A_\phi \hat{\boldsymbol{a}}_{\boldsymbol{\theta}}$$

$$= 4.982\ \hat{\boldsymbol{a}}_{\boldsymbol{r}} +\ 0.566\ \hat{\boldsymbol{a}}_{\boldsymbol{\theta}} +\ 1.964\ \hat{\boldsymbol{a}}_{\boldsymbol{\theta}}$$

(see Figure 22).

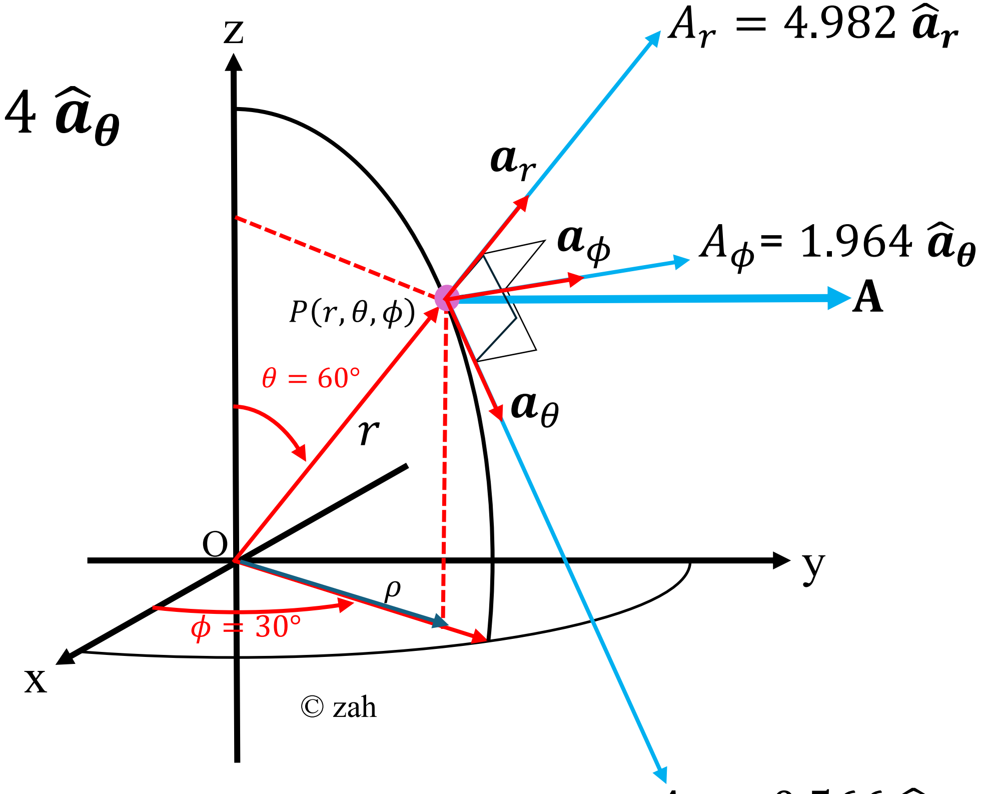


Figure 22

4. Quick consistency check (magnitudes)

The Cartesian magnitude is

$$\|\mathbf{A}\| = \sqrt{3^2 + 4^2 + 2^2} = \sqrt{29} \approx 5.3851648071.$$

Because $\{\hat{\boldsymbol{a}}_{\boldsymbol{r}}, \hat{\boldsymbol{a}}_{\boldsymbol{\theta}}, \hat{\boldsymbol{a}}_{\boldsymbol{\phi}}\}$ are orthonormal, the spherical-component magnitudes must satisfy

$$A_r^2 + A_\theta^2 + A_\phi^2 = \|\mathbf{A}\|^2.$$

Using the numbers above one finds (rounding shown)

$$4.9820508^2 + 0.5669873^2 + 1.9641016^2 \approx 29.000 \ldots$$

so the decomposition is consistent.

Worked Example 5 – General Vector Conversion Using Matrix Form

Given:

Transform vector $\mathbf{A} = 4\hat{\mathbf{i}} - 2\hat{\mathbf{j}} - 4\hat{\mathbf{k}}$ into spherical coordinates at $(-2, -3, 4)$.

(see Figure 23)

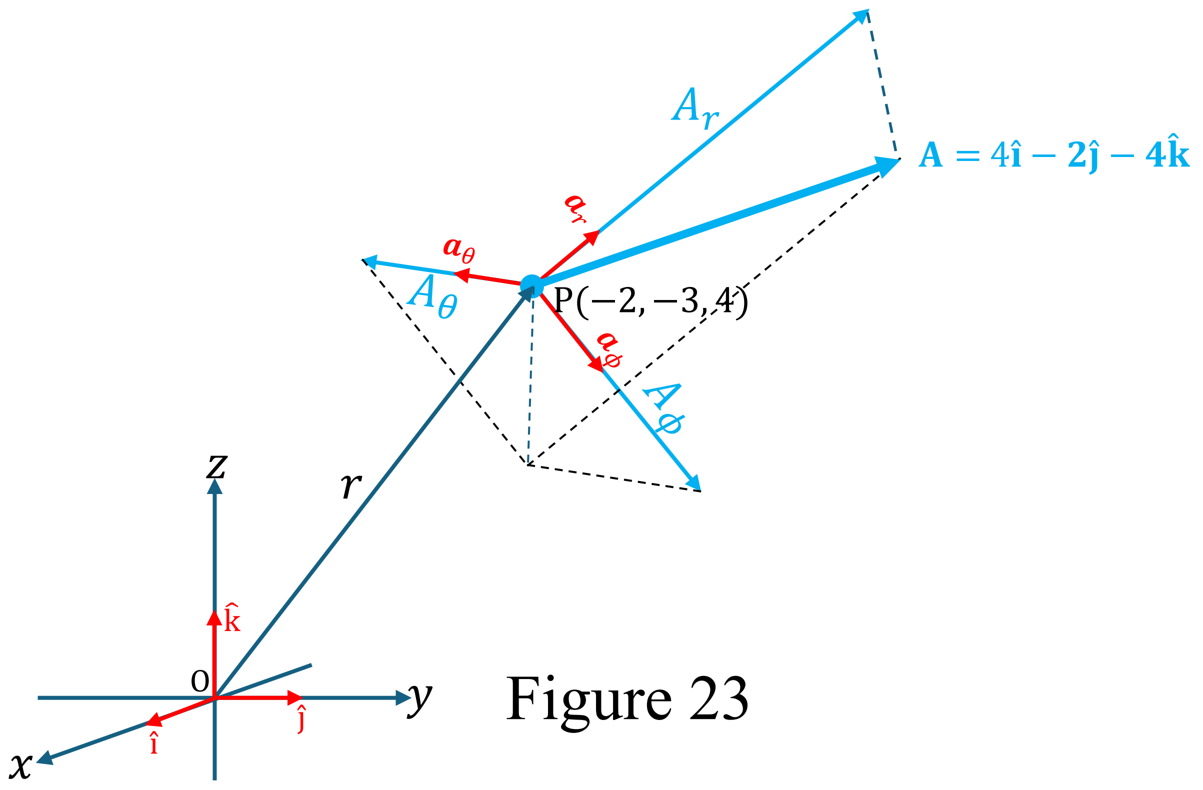


Figure 23

Solution

Step 1: Substitute the given rectangular components

$$x = -2, \qquad y = -3, \qquad z = 4$$

Step 2: Compute the radial distance r

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$r = \sqrt{(-2)^2 + (-3)^2 + 4^2} = \sqrt{4 + 9 + 16} = \sqrt{29}$$

$$r = 5.385 \text{ m}$$

Step 3: Compute the azimuthal angle ϕ

$$\phi = tan^{-1}\left(\frac{-3}{-2}\right) = tan^{-1}\left(\frac{3}{2}\right)$$

$$\phi = tan^{-1}\left(\frac{3}{2}\right) = 56.30°$$

As both x and y are negative, then $\phi = 180° + 56.30° = 236.30°$ (see Figure 24).

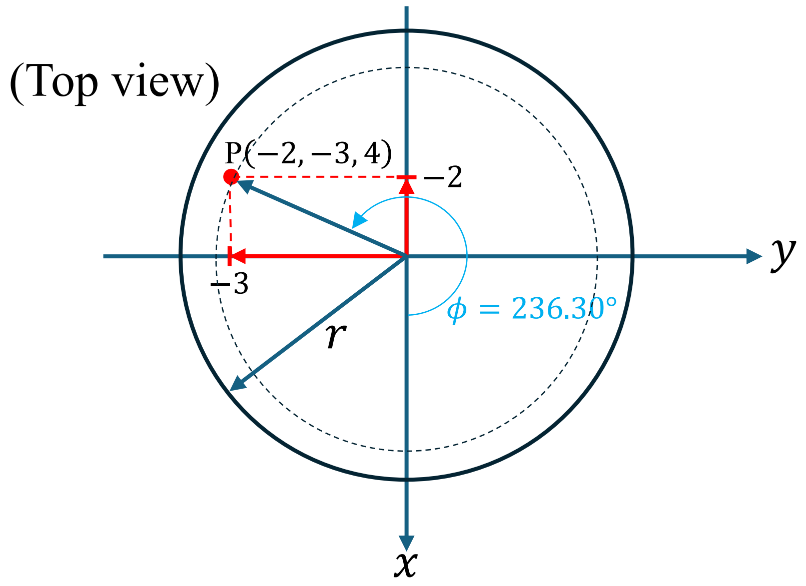


Figure 24

Step 4: Compute the polar angle θ

$$\theta = cos^{-1}\left(\frac{z}{r}\right) = cos^{-1}\left(\frac{4}{5.385}\right) = cos^{-1}(0.802)$$

$$\theta = cos^{-1}(0.802) = 42.029°$$

$$\theta = 42.029°$$

(see Figure 25)

(Front view)

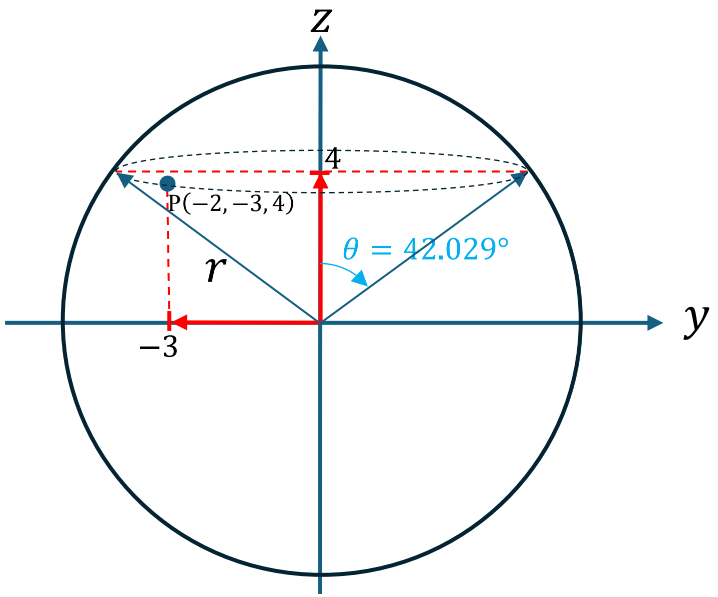


Figure 25

Step 5: Determine local angles

$$r = 3.742, \qquad \theta = 42.029°, \qquad \phi = 236.30°.$$

Step 6: Form the transformation matrix

$$[T] = \begin{bmatrix} \sin\theta\cos\phi & \sin\theta\sin\phi & \cos\theta \\ \cos\theta\cos\phi & \cos\theta\sin\phi & -\sin\theta \\ -\sin\phi & \cos\phi & 0 \end{bmatrix}$$

$$[\mathrm{T}] = \begin{bmatrix} \sin(42.029°)\cdot\cos(236.30°) & \sin(42.029°)\sin(236.30°) & \cos(42.029°) \\ \cos(42.029°)\cos(236.30°) & \cos(42.029°)\sin(236.30°) & -\sin(42.029°) \\ -\sin(236.30°) & \cos(236.30°) & 0 \end{bmatrix}$$

$$[\mathrm{T}] = \begin{bmatrix} -0.371 & -0.557 & 0.743 \\ -0.412 & -0.618 & -0.67 \\ 0.836 & -0.549 & 0 \end{bmatrix}$$

Step 7: Multiply to obtain spherical components

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = [\mathrm{T}] \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} -0.371 & -0.557 & 0.743 \\ -0.412 & -0.618 & -0.67 \\ 0.836 & -0.549 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} -3.343 \\ 2.285 \\ 4.441 \end{bmatrix}.$$

Therefore,

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} -3.343 \\ 2.285 \\ 4.441 \end{bmatrix}.$$

or

$$\mathbf{A} = -3.343\hat{\boldsymbol{a}}_r + 2.285\hat{\boldsymbol{a}}_{\boldsymbol{\theta}} + 4.441\hat{\boldsymbol{a}}_\phi \ \blacksquare$$

Worked Example 6

Given point P$(-2, 6, 3)$ and vector

$$\mathbf{A} = y\hat{\mathbf{i}} + (x+z)\hat{\mathbf{j}}$$

express P and **A** in spherical coordinates.

Evaluate **A** at P in the spherical coordinate system.

(see Figure 26)

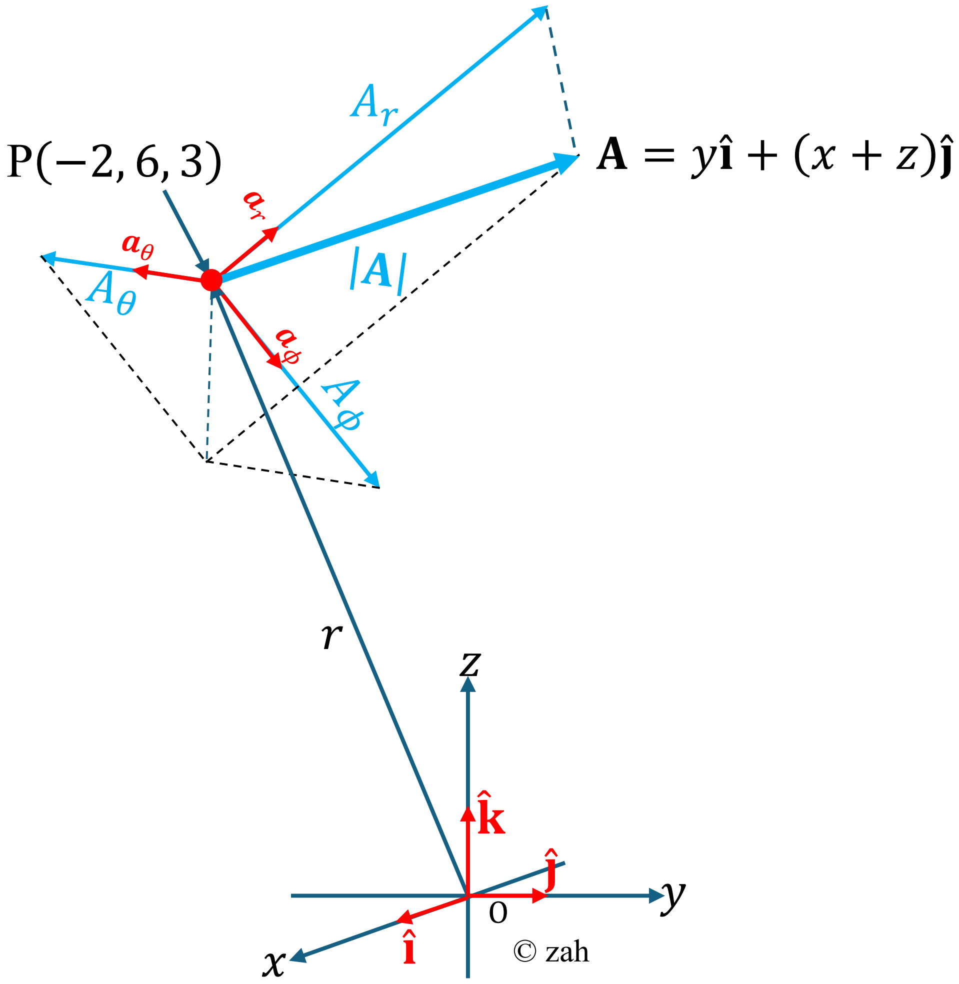


Figure 26

Solution

Step 1: Compute Cartesian vector at P

The general Cartesian component vector for **A** is

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} y \\ x+z \\ 0 \end{bmatrix}$$

Evaluate at $P(-2,6,3)$:

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}_P = \begin{bmatrix} 6 \\ (-2)+3 \\ 0 \end{bmatrix} = \begin{bmatrix} 6 \\ 1 \\ 0 \end{bmatrix}$$

(see Figure 27)

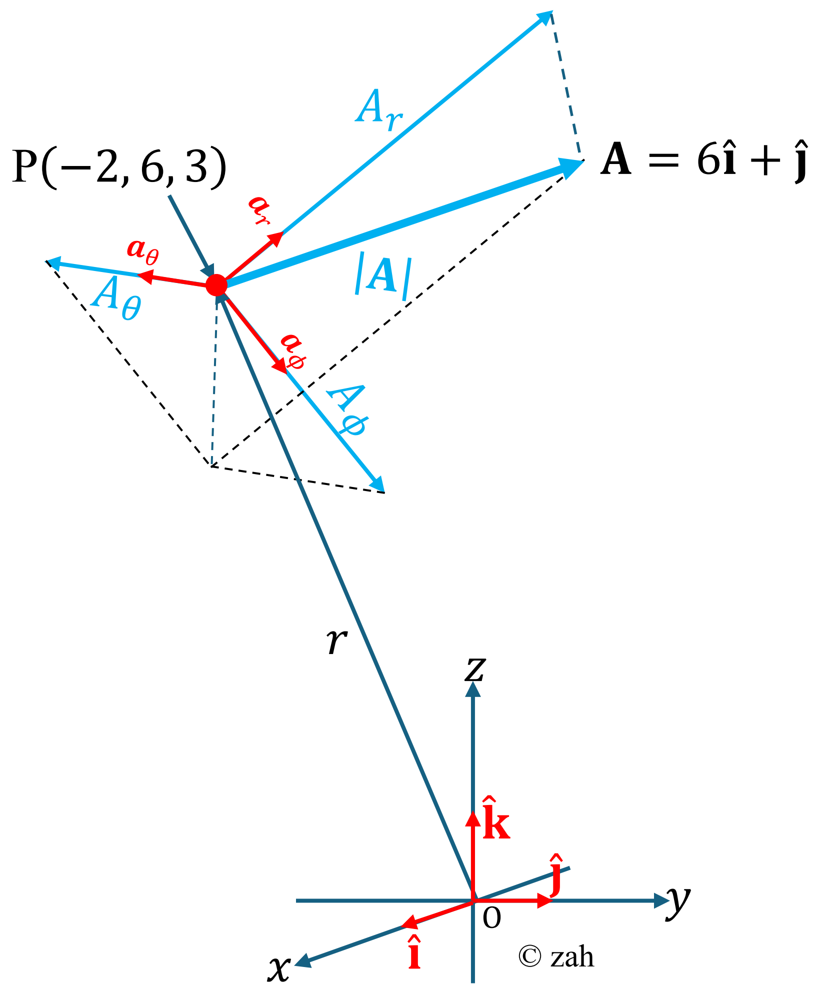


Figure 27

Step 2: Compute position of P in spherical coordinates

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{(-2)^2 + 6^2 + 3^2}$$

$$= \sqrt{4 + 36 + 9} = \sqrt{49} = 7.$$

$$\phi = tan^{-1}\left(\frac{-2}{6}\right) = tan^{-1}\left(-\frac{1}{3}\right)$$

$$\phi = tan^{-1}\left(-\frac{1}{3}\right) = -18.435°$$

As x negative and y is positive, then

$$\phi = 90° + 18.435° = 108.43°$$

(see Figure 28).

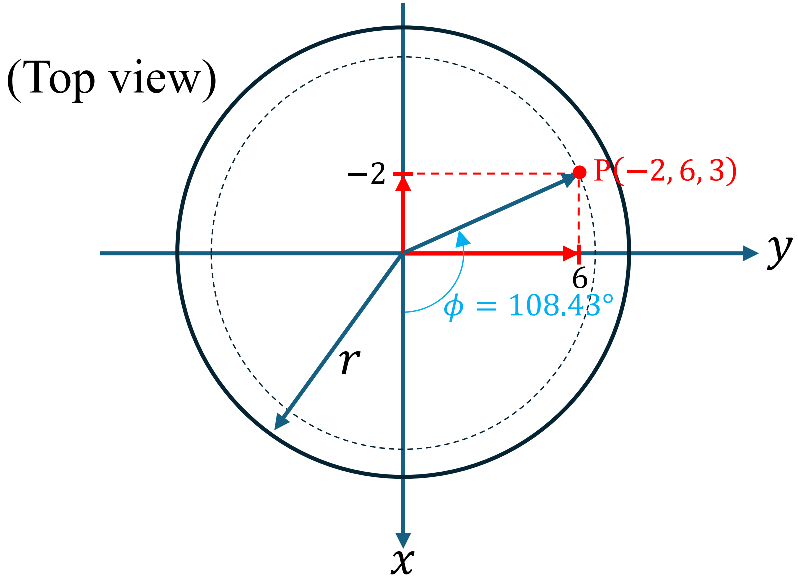


Figure 28

Referring to Figure 29, we get

$$\theta = cos^{-1}\left(\frac{z}{r}\right) = cos^{-1}\left(\frac{3}{7}\right) = tan^{-1}(2)$$

$$\theta = cos^{-1}\left(\frac{3}{7}\right) = 64.62°$$

$$\theta = 64.62°$$

(Front view)

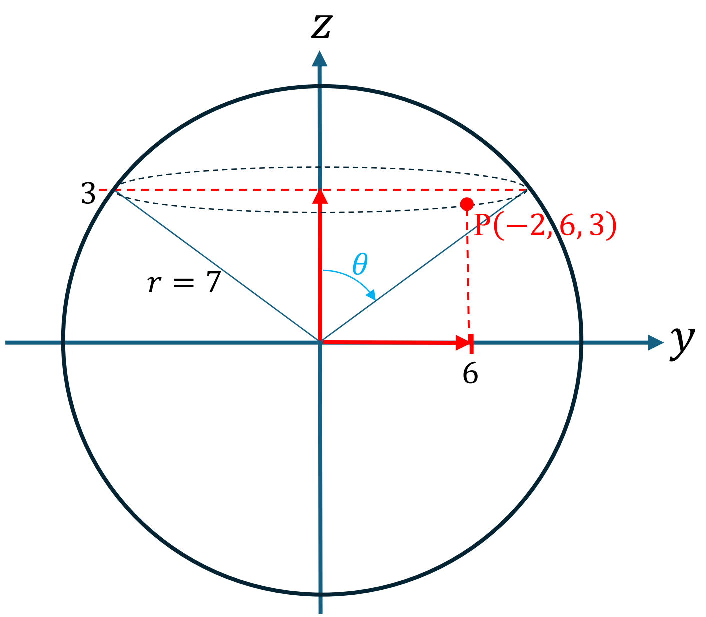


Figure 29

Step 3: Form the transformation matrix

$$[T] = \begin{bmatrix} \sin\theta\cos\phi & \sin\theta\sin\phi & \cos\theta \\ \cos\theta\cos\phi & \cos\theta\sin\phi & -\sin\theta \\ -\sin\phi & \cos\phi & 0 \end{bmatrix}$$

$$[\mathrm{T}] = \begin{bmatrix} \sin(64.62°)\cdot\cos(108.43°) & \sin(64.62°)\sin(108.43°) & \cos(64.62°) \\ \cos(64.62°)\cos(108.43°) & \cos(64.62°)\sin(108.43°) & -\sin(64.62°) \\ -\sin(108.43°) & \cos(108.43°) & 0 \end{bmatrix}$$

$$[\mathrm{T}] = \begin{bmatrix} -0.283 & 0.849 & 0.447 \\ -0.141 & 0.424 & -0.894 \\ -0.949 & -0.316 & 0 \end{bmatrix}$$

Step 4: Multiply to obtain spherical components

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = [\mathrm{T}] \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} -0.283 & 0.849 & 0.447 \\ -0.141 & 0.424 & -0.894 \\ -0.949 & -0.316 & 0 \end{bmatrix} \begin{bmatrix} 6 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -3.343 \\ 2.285 \\ 4.441 \end{bmatrix}.$$

Therefore,

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} -0.848 \\ -0.424 \\ -6.008 \end{bmatrix}.$$

or

$$\mathbf{A} = -0.848\hat{\boldsymbol{a}}_r - 0.424\hat{\boldsymbol{a}}_{\boldsymbol{\theta}} - 6.01\hat{\boldsymbol{a}}_\phi \ \blacksquare$$

Worked Example 7

Express vector

$$\boldsymbol{B} = \frac{10}{r}\boldsymbol{a}_r + r\cos\theta\,\boldsymbol{a}_\theta + \boldsymbol{a}_\phi$$

in Cartesian coordinates. Find $\mathbf{B}(-3, 4, 0)$ and $\mathbf{B}\left(5, \frac{\pi}{2}, -2\right)$.

(see Figure 30)

Solution

Step 1: Write the spherical component vector

The general spherical component vector of **B** is:

$$\begin{bmatrix} B_r \\ B_\theta \\ B_\phi \end{bmatrix} = \begin{bmatrix} \dfrac{10}{r} \\ r\cos\theta \\ 1 \end{bmatrix}$$

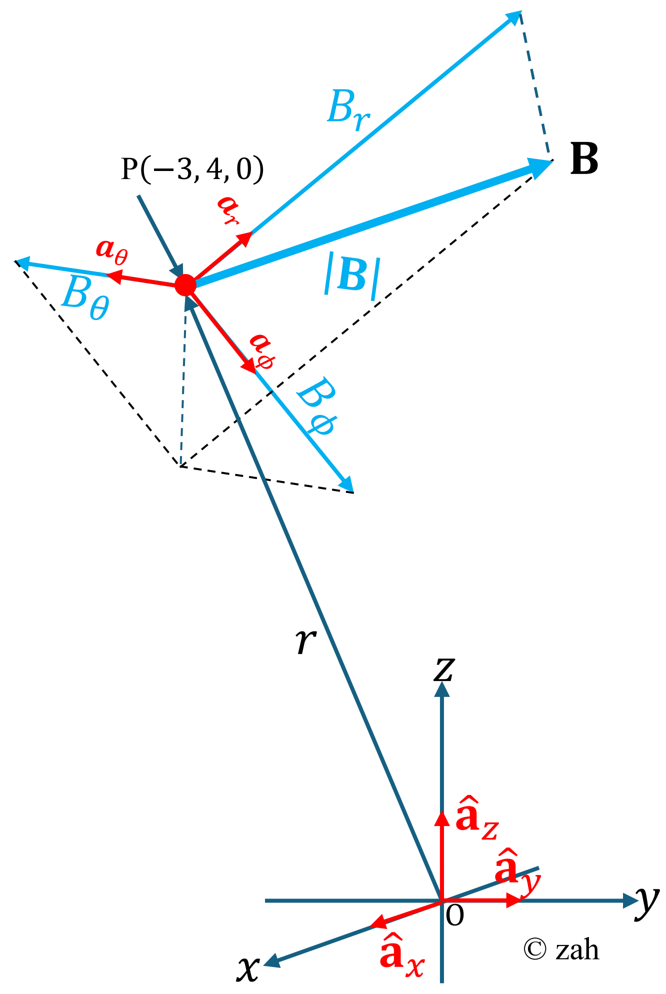


Figure 30

Step 2: Compute the spherical coordinates at $(-3, 4, 0)$:

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{(-3)^2 + 4^2 + 0^2}$$

$$= 5$$

$$\theta = cos^{-1}\left(\frac{z}{r}\right) = cos^{-1}\left(\frac{0}{5}\right) = 90°$$

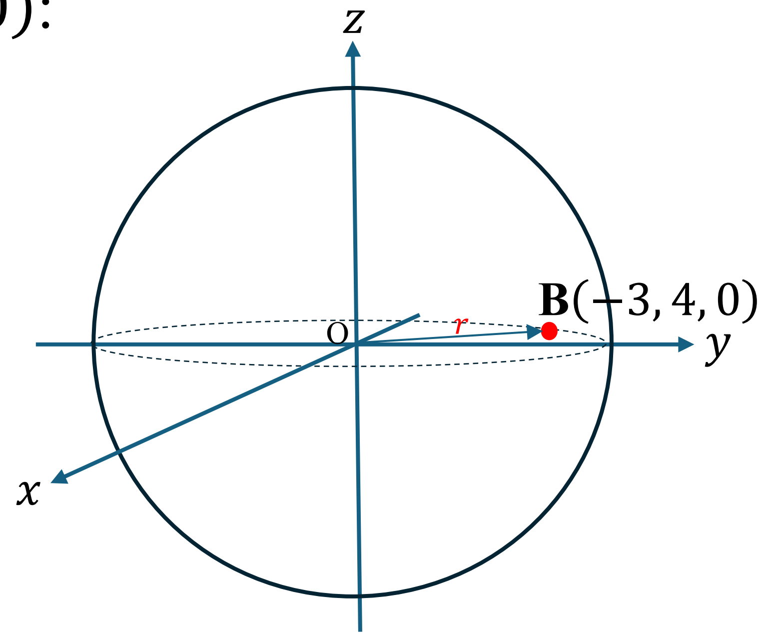
$$\phi = tan^{-1}\left(\frac{y}{x}\right) = tan^{-1}\left(\frac{4}{-3}\right) = -53.13°$$

As x negative and y is positive (see Figure 31),

$$\phi = 180° - 53.13° = 126.87°$$

Hence, the spherical coordinates of the point are:

$$(r, \theta, \phi) = (5, 90°, 126.87°)$$



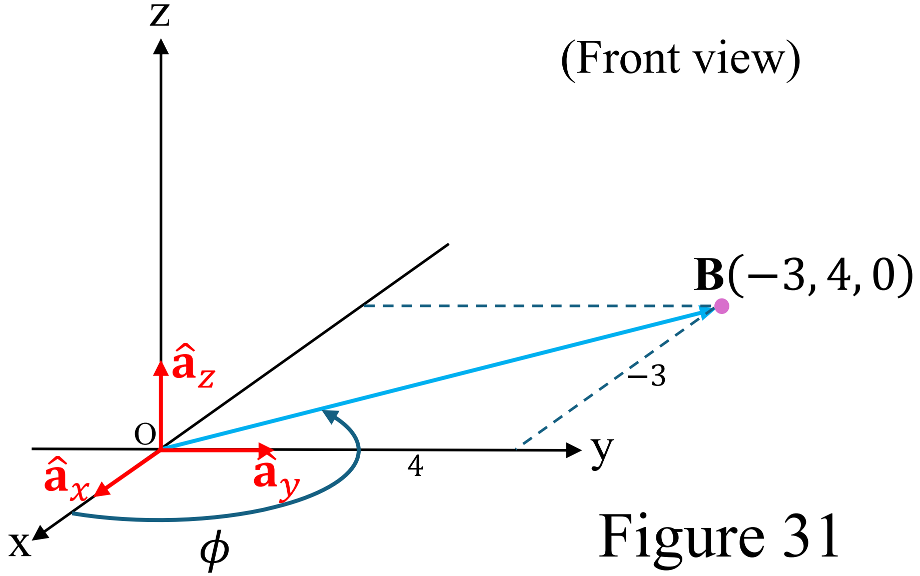


Figure 31

3. Step — Evaluate the spherical components at $(r, \theta, \phi) = (5, 90°, 161.565°)$

$$\begin{bmatrix} B_r \\ B_\theta \\ B_\phi \end{bmatrix} = \begin{bmatrix} \dfrac{10}{r} \\ r \cos \theta \\ 1 \end{bmatrix} = \begin{bmatrix} \dfrac{10}{5} \\ 5 \cos 90° \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

Step 4: Form the spherical-to-Cartesian transformation matrix

$$[T] = \begin{bmatrix} \sin\theta\cos\phi & \sin\theta\sin\phi & \cos\theta \\ \cos\theta\cos\phi & \cos\theta\sin\phi & -\sin\theta \\ -\sin\phi & \cos\phi & 0 \end{bmatrix}$$

Substitute $\theta = 90°$, $\phi = 126.87°$:

$$[\mathrm{T}] = \begin{bmatrix} \sin(90°)\cdot\cos(126.87°) & \sin(90°)\sin(126.87°) & \cos(90°) \\ \cos(90°)\cos(126.87°) & \cos(90°)\sin(126.87°) & -\sin(90°) \\ -\sin(126.87°) & \cos(126.87°) & 0 \end{bmatrix}$$

$$[\mathrm{T}] = \begin{bmatrix} -0.6 & 0.8 & 0 \\ 0 & 0 & -1 \\ -0.8 & -0.6 & 0 \end{bmatrix}$$

Step 4: Form the spherical-to-Cartesian transformation matrix

$$[T] = \begin{bmatrix} \sin\theta\cos\phi & \sin\theta\sin\phi & \cos\theta \\ \cos\theta\cos\phi & \cos\theta\sin\phi & -\sin\theta \\ -\sin\phi & \cos\phi & 0 \end{bmatrix}$$

Substitute $\theta = 90°$, $\phi = 126.87°$:

$$[\mathrm{T}] = \begin{bmatrix} \sin(90°)\cdot\cos(126.87°) & \sin(90°)\sin(126.87°) & \cos(90°) \\ \cos(90°)\cos(126.87°) & \cos(90°)\sin(126.87°) & -\sin(90°) \\ -\sin(126.87°) & \cos(126.87°) & 0 \end{bmatrix}$$

$$[\mathrm{T}] = \begin{bmatrix} -0.6 & 0.8 & 0 \\ 0 & 0 & -1 \\ -0.8 & -0.6 & 0 \end{bmatrix}$$

Step 5: Compute the Cartesian-to-Spherical transformation matrix $[\boldsymbol{T}]^{\boldsymbol{T}}$

$$[\mathrm{T}] = \begin{bmatrix} -0.6 & 0.8 & 0 \\ 0 & 0 & -1 \\ -0.8 & -0.6 & 0 \end{bmatrix}$$

The Cartesian-to-Spherical transformation matrix is

$$[\boldsymbol{T}]^{\boldsymbol{T}} = \begin{bmatrix} -0.6 & 0.8 & 0 \\ 0 & 0 & -1 \\ -0.8 & -0.6 & 0 \end{bmatrix}^{-1}$$

$$[\boldsymbol{T}]^{\boldsymbol{T}} = \begin{bmatrix} -0.6 & 0 & -0.8 \\ 0.8 & 0 & -0.6 \\ 0 & -1 & 0 \end{bmatrix}$$

Step 6: Multiply to obtain Cartesian components

$$\begin{bmatrix} B_x \\ B_y \\ B_z \end{bmatrix} = [\mathrm{T}]^T \begin{bmatrix} B_r \\ B_\theta \\ B_\phi \end{bmatrix}$$

$$[\mathrm{T}]^T = \begin{bmatrix} -0.6 & 0 & -0.8 \\ 0.8 & 0 & -0.6 \\ 0 & -1 & 0 \end{bmatrix}, \qquad \begin{bmatrix} B_r \\ B_\theta \\ B_\phi \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} B_x \\ B_y \\ B_z \end{bmatrix} = \begin{bmatrix} -0.6 & 0 & -0.8 \\ 0.8 & 0 & -0.6 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -2.0 \\ 1.00 \\ 0 \end{bmatrix}$$

Final Answer:

$$\mathbf{B}(-3, 4, 0) = -2.0\hat{\mathbf{a}}_x + 1.0\hat{\mathbf{a}}_y + 0\hat{\mathbf{a}}_z \ \blacksquare$$

Worked Example 8

Given the rectangular vector

$$\mathbf{A} = \frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}} \boldsymbol{a}_x - \frac{yz}{\sqrt{x^2 + y^2 + z^2}} \boldsymbol{a}_z$$

(a) transform the vector **A** into spherical coordinates.

(b) Transform the rectangular coordinate point $P(1, 3, 5)$ into spherical coordinates.

(c) Evaluate the vector **A** at P in spherical coordinates.

Solution

Step 1: Given vector (rectangular form)

$$\mathbf{A} = \frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}}\,\hat{\mathbf{a}}_x - \frac{yz}{\sqrt{x^2 + y^2 + z^2}}\,\hat{\mathbf{a}}_z.$$

Define

$$\rho = \sqrt{x^2 + y^2} \qquad \text{and} \qquad r = \sqrt{x^2 + y^2 + z^2}\,.$$

$$A_x = \frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}} = \frac{\rho}{r}, \qquad A_y = 0, \qquad A_z = -\frac{yz}{\sqrt{x^2 + y^2 + z^2}} = -\frac{yz}{r}.$$

Then

$$\mathbf{A}(x,y,z) = \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \dfrac{\rho}{r} \\ 0 \\ -\dfrac{yz}{r} \end{bmatrix}$$

$$\mathbf{A}(r,\theta,\phi) = [T]\mathbf{A}(x,y,z)$$

Thus

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \sin\theta\sin\phi & \cos\theta \\ \cos\theta\cos\phi & \cos\theta\sin\phi & -\sin\theta \\ -\sin\phi & \cos\phi & 0 \end{bmatrix} \begin{bmatrix} \dfrac{\rho}{r} \\ 0 \\ -\dfrac{yz}{r} \end{bmatrix}$$

Step 2: Transform the rectangular coordinate point $P(1,3,5)$ into spherical coordinates.

At point $P(1,3,5)$,

$$\rho = \sqrt{x^2 + y^2} = \sqrt{1^2 + 3^2} = \sqrt{10}$$

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{1^2 + 3^2 + 5^2} = \sqrt{35} \approx 5.916$$

Therefore,

$$\mathbf{A} = \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \dfrac{\rho}{r} \\ 0 \\ -\dfrac{yz}{r} \end{bmatrix} = \begin{bmatrix} \dfrac{\sqrt{10}}{\sqrt{35}} \\ 0 \\ -\dfrac{(3)(5)}{\sqrt{35}} \end{bmatrix} = \begin{bmatrix} \sqrt{\dfrac{2}{7}} \\ 0 \\ -\dfrac{15}{\sqrt{35}} \end{bmatrix}$$

Step 3: Compute the spherical coordinates at $(1, 3, 5)$:

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{1^2 + 3^2 + 5^2} = \sqrt{35}$$

$$\theta = cos^{-1}\left(\frac{z}{r}\right) = cos^{-1}\left(\frac{5}{\sqrt{35}}\right) = 32.312°$$

Since the point lies in the **first octant**,

the polar angle θ remains $32.31°$.

$$\phi = tan^{-1}\left(\frac{y}{x}\right) = tan^{-1}\left(\frac{3}{1}\right) = 71.565°$$

Hence, the spherical coordinates of the point are:

$$(r, \theta, \phi) = \left(\sqrt{35}, 32.312°, 71.565°\right)$$

(see Figure 32)

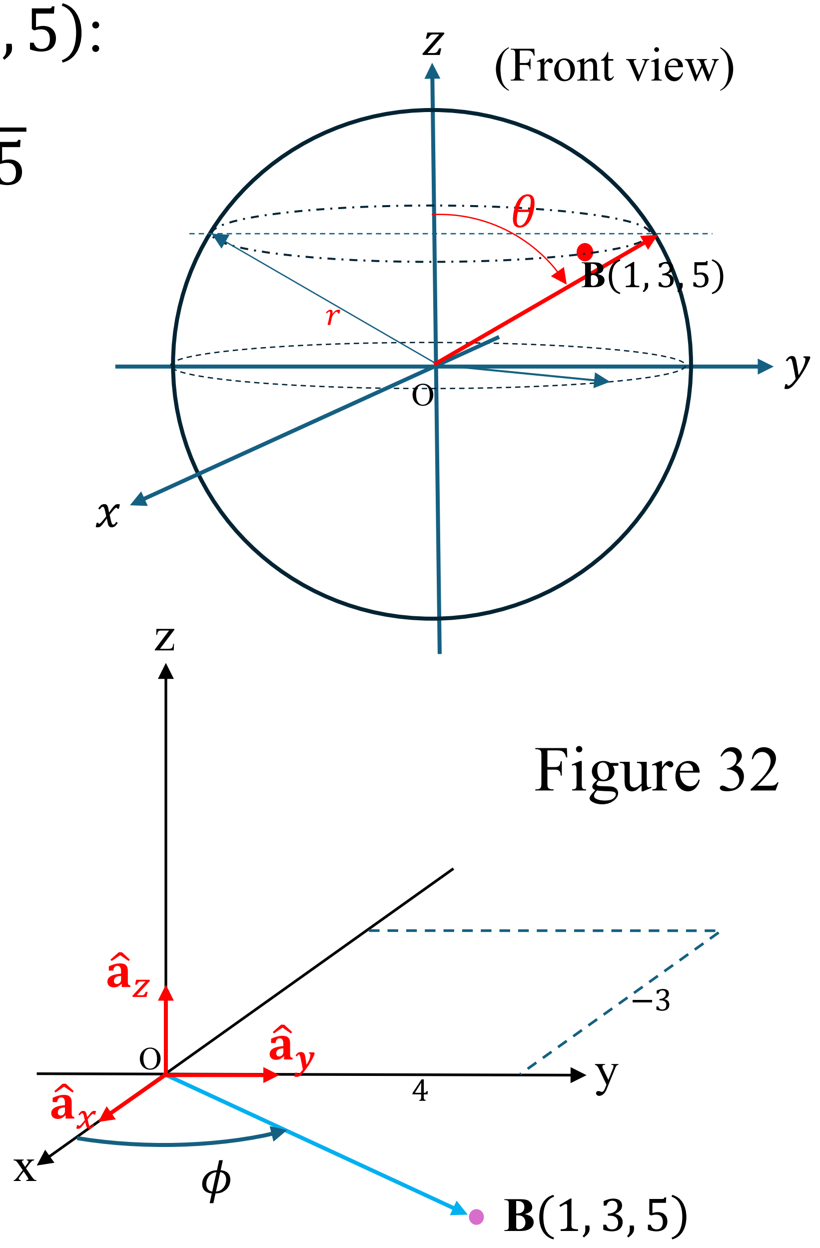


Figure 32

Step 4: Evaluate **A** at $P(1,3,5)$

At $P(1,3,5)$

$$\rho = \sqrt{10}, \qquad (r,\theta,\phi) = \left(\sqrt{35}, 32.312°, 71.565°\right)$$

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \sin\theta\sin\phi & \cos\theta \\ \cos\theta\cos\phi & \cos\theta\sin\phi & -\sin\theta \\ -\sin\phi & \cos\phi & 0 \end{bmatrix} \begin{bmatrix} \frac{\rho}{r} \\ 0 \\ -\frac{yz}{r} \end{bmatrix}$$

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin(32.312°)\cos\phi & \sin(32.312°)\sin\phi & \cos(32.312°) \\ \cos(32.312°)\cos\phi & \cos(32.312°)\sin\phi & -\sin(32.312°) \\ -\sin\phi & \cos\phi & 0 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{10}}{\sqrt{35}} \\ 0 \\ -\frac{(3)(5)}{\sqrt{35}} \end{bmatrix}$$

Compute the spherical components of A:

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} 0.169 & 0.508 & 0.845 \\ 0.267 & 0.803 & -0.535 \\ -0.950 & 0.316 & 0 \end{bmatrix} \begin{bmatrix} 0.5345 \\ 0.0 \\ -2.534 \end{bmatrix}$$

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} -2.051 \\ 1.499 \\ -0.508 \end{bmatrix}$$

Hence, the spherical form of the vector **A** is:

$$\mathbf{A} = -2.051\hat{\mathbf{a}}_r + 1.499\hat{\mathbf{a}}_\theta - 0.508\hat{\boldsymbol{a}}_\phi \blacksquare$$

END OF UNIT 4